

Incidence structure and solver limits of contact Delassus operators

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Abstract

The Delassus operator couples contact impulses through the bodies they share. We express this coupling as a bodywise incidence assembly with inverse spatial inertia as its block conductance. In common patch coordinates, a unique contact per body pair yields a block Laplacian with a possibly zero Dirichlet term precisely when dynamic bodies have at most two incident patches with opposite signs. Branching violates that sign condition without invalidating the assembly or its spectral correspondence with a vertex pencil. Patch wrenches remove internal corner-force self-stress and preserve the frictionless normal and moment cone; a tangential spin ellipsoid changes the contact law. These distinctions identify where graph structure gives an exact linear solve and where it does not predict nonlinear convergence. Chain elimination reproduces dense contact updates, but a fixed spectral penalty fails on most scenes in a parameter sweep. Residual balancing lowers the median paired ratio of capped iteration counts under the measured de Saxcé refresh schedule, while losing on some scenes and incurring additional factorizations. A shift ablation reverses the penalty ordering on the dense random system. On the same operators we measure the stopping convention directly: a fixed physical tolerance is unit-consistent, while a genuinely dimensionless stop and an absolute tolerance differ from it, and the delivered absolute tolerance is unit-dependent. The resulting description concerns contact conditioning and linear structure, rather than a universal solver or a spectral warning of physical failure. We state the relation to the closest splitting architecture for the same exact Coulomb law, whose outer map and stopping metric differ from ours.

1. Introduction

Repeated contact calculations enter both trajectory optimization and differentiation of robot dynamics. Contact constraints can be optimized together with a trajectory [1]. Analytical dynamics derivatives exploit the algebraic structure of the equations [2]. In an impulse formulation, eliminating body velocities leaves the Delassus operator, which maps contact impulses to changes in contact velocity [3,4]. A light body between heavy neighbours can make this map difficult for local iterations even in a simple stack. The operator describes this coupling, but convergence also depends on the contact law and its discretization.

For fixed geometry, we consider

$$\begin{aligned} u &= G\lambda + b, \quad G = JM^{-1}J^T, \\ K_\mu &\ni \lambda \perp u + \Gamma(u) \in K_\mu^*, \\ K_\mu &= \prod_C \{ (\lambda_n, \lambda_t) : \lambda_n \geq 0, \|\lambda_t\| \leq \mu_C \lambda_n \}, \\ \Gamma_C(u_C) &= (\mu_C \|u_{t,C}\|, 0, 0). \end{aligned}$$

Here b is the free contact velocity including the prescribed bias, and Γ is the de Saxcé correction [5–9]. Dropping this correction gives an associated convex problem. We use *exact Coulomb* to distinguish the corrected contact law from that relaxation; the term does not imply exact arithmetic or convergence of every iteration. A diagonal compliance is stated separately because it changes the problem, whereas the ADMM penalty changes how a fixed problem is solved.

The contact graph has bodies as vertices and contacts as edges. The coupling graph has contacts as vertices, with connections generated by shared dynamic bodies. A shared static support contributes no coupling through the inverse mass. For point masses with parallel normals, inverse mass supplies a scalar conductance. Rigid rotation requires a block conductance. A face

represented by corner impulses introduces a further issue: different corner forces can produce the same body wrench. We characterize the incidence signs in common coordinates and then distinguish elimination of those redundant force coordinates from approximation of their admissible cone.

Contributions

Several ingredients are known algebra and known statics, and we state them in contact coordinates because they fix the scope of what follows. The incidence Gram identity of Lemma 1 is the standard factorization of a projected inverse inertia. Its off-diagonal sign condition is the orientation question for the signed contact line graph, whose balance criterion is Harary’s [20,21]: under the common-gauge and unique-body-pair hypotheses the off-diagonal blocks are negative semidefinite exactly when every dynamic body has degree at most two with opposite incidences, while a branching body admits no orientation because two signs must coincide. The equality between the nonnegative-normal-cone image and the centre-of-pressure condition of a rectangle is classical support statics. The vertex-pencil identity between the two Gram matrices of the Jacobian and the graph operator holds for any Jacobian and any positive definite mass matrix, and is used here as an exact reference rather than claimed as a new spectral theorem. Block elimination is related to the Schur complements used in articulated-body dynamics [10,11].

The contact-specific content is the common-coordinate assembly with inverse spatial inertia as block conductance, its exact reduction to a block Laplacian with a possibly zero Dirichlet term under those hypotheses, the degree-three counterexample, and the wrench quotient that separates removal of internal corner-force self-stress from approximation of the admissible tangential cone. For the spatial block assembly the off-diagonal weight is the positive definite block $W_k = \text{Ad}_k M_k^{-1} \text{Ad}_k^T$ of the shared body, which makes the spatial assembly a matrix-weighted incidence Gram assembly. Off-diagonal blocks are negative semidefinite, and the assembly is the Laplacian L_H of Theorem 2, only under the stated unique-body-pair, common-frame and incidence-sign assumptions; without them the signed assembly and its positive semidefiniteness remain valid. It is not the graph connection Laplacian with orthogonal transport: Ad_k is block unit-triangular and not orthogonal at nonzero displacement, and W_k is not a scalar times an orthogonal matrix. The orthogonal-connection reading applies only to the scalar friction-row and point-mass specialisation of §2.2, where the relative rotation is orthogonal.

For an isotropic sliding contact we give an explicit local sliding lemma. With $G = gI$, Euclidean cone projection, and the de Saxcé correction refreshed every iteration, the linearised iteration at the sliding fixed point has the spectrum of Proposition 6, whose spectral radius is minimised at

$$\rho = g/(1 + \mu^2).$$

The constant depends on friction as well as on the isotropic Delassus scale, and the frozen-correction crossing is $\rho = g$. A bounded search did not locate this constant in prior penalty-selection literature; this is a null result of that search, not a priority claim. The load experiment compares slow-ramp sliding thresholds with the static directional radius of the admissible load section on flat supports. It measures finite-rate agreement, while a polytopal representation requires polygonal friction. These statements have different hypotheses and do not supply a universal penalty or failure predictor.

The negative results delimit those claims. The structural spectral rule meets its parameter-distance criterion on only five of the tested scenes. The sweep contains twenty-one scenes. Adaptive penalties improve the median paired ratio of recorded iteration counts, but the shift ablation and failed runs prevent a general superiority claim. Separate linear-system and external-solver experiments test the algebra and contact law; their timings are not pooled. The complete compliant-to-rigid solver of Carpentier and colleagues [12] is not evaluated here.

2. Incidence and patch coordinates

2.1 Bodywise assembly

Independent body inertias let us separate the contribution of each shared body. Let M be block diagonal with positive definite blocks for dynamic bodies, and omit static bodies from its inverse. Let P_k project the global velocity onto body coordinates, and let s_k collect the incidence signs at body k . At contact c , write the Jacobian block as a signed sum of the kinematic maps of the incident bodies. The signs encode the chosen ordering of the body pair; the kinematic maps encode frames, moment arms, and the measured velocity components. Two contacts interact only through bodies incident to both of them. Expanding the mass inverse therefore gives the following assembly identity.

Lemma 1 (bodywise assembly). With contact-body signs σ_{ck} , kinematic maps B_{ck} , and dynamic body inertias M_k ,

$$G_{cc'} = \sum_k \sigma_{ck} \sigma_{c'k} B_{ck} M_k^{-1} B_{c'k}^T.$$

Proof. Substitute $J_c = \sum_k \sigma_{ck} B_{ck} P_k$ into $J_c M^{-1} J_{c'}^T$. The block diagonal inverse gives

$$P_k M^{-1} P_k^T = \delta_{kl} M_k^{-1}.$$

All terms involving distinct bodies therefore vanish. The remaining terms are precisely those in the displayed sum. If distinct contacts share at most one body, their off-diagonal coupling contains at most one such term. ■

For patch twists, all incident kinematic maps can be transported to a common world point and frame. In these coordinates the map associated with a body is independent of the particular patch, and its inverse spatial inertia becomes the conductance W_k . The body conductance remains positive definite under this transport because the transport is invertible. Here Ad_k transports a body twist to the chosen common point and frame; it is not orthogonal at nonzero displacement, so W_k is not a scalar times an orthogonal matrix. The spatial assembly is therefore a matrix-weighted incidence Gram assembly, not a graph connection Laplacian with orthogonal connection. It is a block Laplacian, with the sign and degree conditions on its off-diagonal blocks, only under the hypotheses of Theorem 2; the incidence Gram form itself holds for arbitrary body degree. Each body contributes an incidence outer product multiplied by a positive definite block. This factorization explains the sparsity and positive semidefiniteness of the contact operator for arbitrary body degree. It does not, by itself, establish the signs required of a Laplacian.

$$\begin{aligned} W_k &= Ad_k M_k^{-1} Ad_k^T, \\ G_{\text{common}} &= \sum_k (s_k s_k^T) \otimes W_k. \end{aligned}$$

2.2 When the assembly is a Laplacian

The sign condition can be stated without referring to a solver. Assume one patch per body pair and use the common coordinates of the preceding subsection. If a dynamic body touches two patches with opposite signs, it contributes a negative conductance to their off-diagonal coupling. A body touching one patch contributes only to that patch's diagonal, giving a Dirichlet term. With these incidences, the operator is a block Laplacian with a possibly zero Dirichlet term. The theorem also identifies exactly why a branching contact graph falls outside this class.

Theorem 2 (Laplacian characterization). Under the common-gauge and unique-body-pair hypotheses, all off-diagonal blocks are negative semidefinite if and only if each dynamic body has degree at most two and its two incidences, when present, are opposite. Under these conditions,

$G=L_H$, where shared bodies supply the block weights and singly incident bodies supply Dirichlet terms. Without these conditions, the signed assembly and positive semidefiniteness remain valid.

Proof sketch. By Lemma 1, an adjacent off-diagonal block is either $+W_k$ or $-W_k$. Since W_k is positive definite, the negative sign is necessary and sufficient. At a body with at least three incidences, two signs must coincide, producing a positive block. When the stated degree condition holds, each incidence outer product is either an edge difference outer product or a diagonal coordinate outer product. Summing these terms gives the Laplacian and its Dirichlet residue. ■

For a branching body with a triplet of incident patches, one possible incidence vector and its contribution are

$$\begin{aligned} s &= (1, 1, -1)^T, \\ (s s^T) \otimes W &= [[W, W, -W], [W, W, -W], [-W, -W, W]], \\ W &> 0. \end{aligned}$$

The positive off-diagonal block is the counterexample at degree three. The obstruction cannot be removed by changing contact orientation. Multiplying contact rows by signs changes the incidence signs but leaves the same impossibility of pairwise opposition at a branching body. Orienting a path successively, or a cycle cyclically, gives opposite signs at each body incident to a pair of patches, regardless of cycle parity. The orientation condition is Harary balance of the signed contact line graph [20,21], and a sign change in contact coordinates must also transform the impulse cone before solving the physical contact problem. This result concerns dynamic body degree; adding a ground contact can change that degree even when the visible arrangement still resembles a ring. A ground-free ring can pass the block sign test, whereas adding a ground contact at a body already incident to a pair of patches creates the branching obstruction. Appendix A checks the assembly, rank, and spectral identities on chains and a lattice.

The scalar specialization has additional geometric restrictions. With translation only and parallel normals, the conductance is inverse mass and the kinematic maps reduce to signed scalar incidence. If the normals differ, the off-diagonal also contains their dot product. Its sign can then change even when the incidence signs are opposite, so the scalar angle condition must accompany the graph condition. Rotation adds a moment-arm term of unrestricted scalar sign. The common-frame block representation absorbs that term into spatial inertia, which is why rotation invalidates the scalar reading without invalidating the block assembly.

Friction rows provide a different test of the same reasoning. For point masses with a common orthonormal contact frame, normal and tangential coordinates produce independent copies of the scalar operator. Let E_c have the contact-frame basis vectors as rows, and let L denote the scalar signed incidence Gram matrix. When contact frames differ, their relative rotation appears between the blocks. In either case, changing the friction coefficient changes the cone while leaving this linear kinematic assembly unchanged. The graph structure therefore does not establish convergence of a frictional projected iteration.

$$\begin{aligned} G_{cc'} &= L_{cc'} E_c E_{c'}^T, \\ E_c &\equiv E = G = L \otimes I_3. \end{aligned}$$

2.3 Coordinates and the spectrum

The common coordinates used to identify the Laplacian need not be the coordinates used to project onto the contact cone. A block diagonal invertible map converts common patch twists to local normalized coordinates, and the Delassus operator transforms by congruence. Congruence preserves rank and positive semidefiniteness, but it does not generally preserve eigenvalues. It also does not preserve the sign of each off-diagonal block when different blocks use different transformations. Consequently, the common-frame sign theorem cannot be applied directly to a

local matrix without carrying the coordinate maps. Any spectral estimate used by a solver must refer to that solver's actual Jacobian.

The spectral correspondence itself requires fewer assumptions than the Laplacian characterization. For the chosen coordinates, define

$$K = JM^{-1/2}.$$

The contact operator and the vertex operator are the two Gram matrices of this same rectangular map. Their nonzero eigenvalues coincide even on a lattice whose incidence signs violate the Laplacian condition. Thus the graph viewpoint remains useful for spectral computation outside the unsigned class. Appendix A includes the lattice in the spectral check.

Theorem 3 (vertex pencil). For any J and positive definite M , the following operators have the same nonzero spectrum with multiplicity:

$$JM^{-1}J^T, M^{-1/2}J^TJM^{-1/2}.$$

Equivalently, the nonzero generalized eigenvalues of $J^TJx = \lambda Mx$ are the nonzero eigenvalues of G . Moreover, $\text{rank } G = \text{rank } J$.

Proof sketch. A nonzero eigenvector of KK^T maps through K^T to an eigenvector of K^TK with the same eigenvalue. The map is injective on that eigenspace, and the reverse map through K gives equal multiplicities. Substitution by the inverse mass square root gives the generalized pencil. Finally, multiplication by an invertible mass square root preserves rank, and a matrix and its Gram matrix have the same rank. ■

The pencil can be applied without forming the contact matrix, using products with the Jacobian, its transpose, and the inverse mass or a mass solve. This avoids contact-matrix assembly, but does not guarantee that a fixed eigensolver budget resolves the smallest positive mode. The penalty experiment uses the spectrum in the coordinates of its own contact solve.

2.4 What patch lumping preserves

Let the rectangular patch have a right-handed orthonormal frame and positive half-extents, defined by

$$(t_1, t_2, n), \quad n = t_1 \times t_2, \quad e_1 > 0, \quad e_2 > 0.$$

Normalize physical moments by their corresponding extent scales to obtain the wrench coordinates:

$$\begin{aligned} \ell_{\text{patch}} &= \sqrt{(e_1^2 + e_2^2)/3}, \\ w &= (F_n, M_{t1}/e_2, M_{t2}/e_1, F_{t1}, F_{t2}, M_n/\ell_{\text{patch}}). \end{aligned}$$

The corresponding twist coordinates are scaled dually to preserve virtual work. In the impulse solve, force symbols denote resultant impulses and moment symbols denote angular impulses divided by the stated length scales. The same cone inequalities apply to static forces and moments in the load analysis. Summing the corner forces and their moments about the patch centroid defines the wrench map C . Its kernel consists of corner-impulse distributions with zero resultant force and zero resultant moment. Quotienting by this kernel removes redundant coordinates while preserving the generalized body impulse.

Lemma 4 (wrench quotient). For a nondegenerate rectangular patch on a free rigid body, the point Jacobian has rank six. It has twelve rows. The wrench map has rank six and satisfies

$$\begin{aligned} J_{\text{points}}^T \lambda &= J_{\text{patch}}^T C \lambda, \\ \ker C &= \ker J_{\text{points}}^T. \end{aligned}$$

Proof sketch. Differences of corner rows generate pure moments about each spatial axis, while the original rows span the resultant force directions. Their row space is therefore the full spatial wrench space. The patch Jacobian block is invertible because its force and moment directions form a basis with nonzero extent scalings. Summing corner forces and moments gives the displayed identity, and injectivity of the transposed patch block identifies the kernels. ■

The quotient also preserves a specific part of the feasible set. For frictionless normal corner forces, the centre of pressure is a convex combination of the corner positions. Conversely, any centre of pressure in the rectangle can be realized by nonnegative corner loads with the prescribed normal resultant. This proves equality of the normal and moment cone image, rather than an inclusion obtained by relaxation. The same construction extends to planar polygons by replacing the rectangle with the convex hull of the vertices. Non-collinearity is needed to retain the full wrench rank when tangential directions are included.

Theorem 5 (normal cone image). In normalized rectangular patch coordinates, the image of nonnegative normal corner impulses is

$$K_{nm} = \{(F, M_1, M_2): F \geq 0, |M_1| \leq F, |M_2| \leq F\}.$$

Proof sketch. Write a corner contribution and the proposed corner loads as

$$\begin{aligned} \lambda(1, \alpha, \beta), \quad \alpha, \beta \in \{-1, +1\}, \\ \lambda_{\alpha\beta} = (F + \alpha M_1)(F + \beta M_2)/(4F), \quad F > 0. \end{aligned}$$

Nonnegativity gives both moment bounds by the triangle inequality. Conversely, the displayed loads are nonnegative and recover the prescribed resultant and moments. For zero resultant, the inequalities force zero moments. In physical coordinates these are precisely the centre-of-pressure inequalities. ■

This exact statement does not extend to the tangential cone used in the patch solver. The linear map retains normal torque, but the chosen Contensou-type ellipsoid has a different spin capacity from the corner Coulomb cones [13]. Pure spin isolates the difference because tangential corner forces can form a torque without a tangential resultant. For the same normal impulse, the capacities are

$$\begin{aligned} \tau_{\text{corner}} &= \mu F_n \sqrt{e_1^2 + e_2^2}, \\ \tau_{\text{patch}} &= \mu F_n \ell_{\text{patch}}, \\ \tau_{\text{corner}}/\tau_{\text{patch}} &= \sqrt{3}. \end{aligned}$$

Pure spin therefore distinguishes a change of contact law from loss of a wrench component.

Local rank restoration does not imply full row rank of the scene Jacobian. Loops can sustain wrench self-stress after corner redundancy is removed, as the lattice in Appendix A demonstrates. If the complete patch Jacobian has full row rank, positive definiteness makes the fixed-shift affine variational inequality coercive and strongly monotone on the closed convex cone. It therefore has a unique solution. A locally constant active set, strict complementarity, and a nonsingular active-set residual Jacobian permit application of the implicit function theorem. These hypotheses neither establish uniqueness of the full Coulomb fixed point nor differentiation through a contact-regime change.

3. Linear consequences

On a patch chain, only adjacent interfaces share a dynamic body. The Delassus matrix is therefore block tridiagonal in chain order, and adding an ADMM diagonal shift preserves that pattern. Eliminating successive diagonal blocks solves the same linear update as a dense factorization. Appendix B compares the resulting outer iterations. The converged cases reproduce the dense updates; the driven mass-ratio column fails with both factorizations. Agreement of the linear solves does not make the outer contact map contractive.

This observation should not be transferred directly to projected iteration. Preconditioning changes the metric of the linear step. When a cone constraint is active, a Euclidean projection need not be the projection associated with that metric. An exact inverse can consequently solve an interior linear problem without yielding a valid single-step constrained update. ADMM separates the linear solve from cone projection, so the block factorization has an unambiguous role there.

Classical algebraic multigrid provides a separate linear baseline. The operator in Appendix C is the scalar incidence Gram matrix with inverse-mass weights, shifted when necessary to obtain a positive definite system. It is the normal Delassus operator only under the translation and frame assumptions stated above. Both multigrid variants reach the linear tolerance with conjugate-gradient acceleration. Their standalone cycles stall on the tested non-M-matrices, while the chain cases converge. This coincidence does not establish a causal consequence of the sign theorem. The experiment evaluates classical multigrid, not the nearly-linear algorithms of Spielman–Teng, Koutis–Miller–Peng, or Kyng–Sachdeva [14–16].

A graph surrogate can also be a poor preconditioner for the actual contact matrix. Appendix D compares the graph, block-Jacobi, and unpreconditioned methods on identical right-hand sides within each scene. The graph construction agrees with the relevant normal block on the mass-ratio chain and reduces Krylov work. On the packing it differs from that block and costs more iterations and more solve time than Jacobi scaling. Those times exclude preconditioner construction, whereas the multigrid times include setup. These cases show a benefit when the surrogate agrees with the tested block and a loss when it does not. They do not isolate operator agreement as the cause of that difference.

4. Penalty selection for a fixed contact problem

The incidence identity permits spectral computation through Jacobian products without assembling the contact matrix. It does not guarantee a prescribed eigenvalue accuracy at a fixed Krylov budget. More fundamentally, accurate eigenvalues do not contain the contact activity or projection derivative that governs a nonlinear iteration. We test the resulting parameter prediction with a dense spectrum so that estimation error cannot explain its failures.

4.1 Sweep and paired comparison

Appendix E uses the same operator for the spectrum, ADMM update, de Saxcé correction, and stopping residual. Write that operator as

$$\begin{aligned} G_\eta &= G_0 + \text{diag}(\eta), \quad G_0 = JM^{-1}J^T, \\ R_\eta(\lambda) &= \|\lambda - \Pi_K[\lambda - D(G_\eta\lambda + b + \Gamma(G_\eta\lambda + b))]\|_\infty, \\ D_c &= \|(G_\eta)_{cc}\|_2^{-1} I. \end{aligned}$$

Appendix I defines the exported scenes and their contact coordinates. The packing and granular-step cases include diagonal compliance; its multiplier is included in the sweep table. The remaining scenes use the rigid operator. Thus the packing spectrum in this experiment is not the unregularized spectrum in the preconditioner comparison. Each penalty candidate starts from zero. The reference penalty is the converged grid point requiring the fewest iterations, with ties resolved toward the smaller penalty. Boundary winners are grid minima, not established optima beyond the sampled range.

The candidate rules and grid are

$$\begin{aligned} \rho_A &= \lambda_{\min}^+(G_\eta), \quad \rho_B = \sqrt{(\lambda_{\min}^+(G_\eta)\lambda_{\max}(G_\eta))}, \\ \rho_{\text{struct}} &= \rho_A \text{ if the structural test passes, and } \rho_B \text{ otherwise,} \\ \rho_j &= \lambda_{\min}^+(G_\eta) 10^{a_j}, \quad a_j \in \text{inspace}(a_{\min}, a_{\max}, N_{\text{grid}}). \end{aligned}$$

The structural test checks the Jacobian support: each represented body has at most two incident contacts, distinct contacts share at most one body, and the Jacobian blocks of paired incidences

are opposite. Its result depends on the represented coordinates; multiple corner contacts count separately before patch reduction. Articulated cases use their generalized operator in the sweep and are not evidence for a bodywise Laplacian decomposition of a coupled mass matrix. Appendix E states the eigenvalue cutoff, grid range, and failure counts alongside each scene.

The structural rule passes the factor-of-two parameter criterion on five scenes. The remaining sixteen fail it. Chains with inactive constraints often favour a boundary grid value, while packings prefer values above the geometric-mean scale. The same spectral choice is therefore not uniformly predictive across regimes. Selecting whichever spectral rule lies nearer the grid winner after the sweep would change the task from prediction to retrospective selection.

Appendix F is the primary algorithmic comparison. It reruns the swept penalty at the same natural-map target used for every fixed and adaptive method in that table. The methods share zero initialization and the ADMM/Cadoux implementation with sparse direct linear solves. The mean initialization is $\text{tr}(G_\eta)/\text{dim}(G_\eta)$. The structural initialization is the declared rule above. The He-Yang-Wang residual-balancing rule increases or decreases the penalty when primal and dual residual norms become imbalanced [17,18]. The normalized variant applies the same decision to relative residuals, following Wohlberg [19]. The implemented residuals are

$$\begin{aligned} r_p &= x - z, \quad r_d = \rho(z - z_{\text{previous}}), \\ r_{p,\text{rel}} &= \|r_p\|_2 / \max(\|x\|_2, \|z\|_2, \varepsilon_{\text{den}}), \\ r_{d,\text{rel}} &= \|r_d\|_2 / \max(\|\gamma\|_2, \varepsilon_{\text{den}}), \quad \varepsilon_{\text{den}} = 10^{-300}. \end{aligned}$$

Here γ is the unscaled dual variable and remains unchanged when the penalty changes. Every penalty change triggers a new sparse factorization, so iteration counts do not measure equal amounts of work.

The measured solver updates the de Saxcé shift between inner ADMM blocks and holds it fixed within each block. It starts with zero shift and refreshes tangential speeds after the inner primal residual meets a numerical threshold. The threshold depends on the change in the shift, with the free-velocity bias used for the first block. Appendix F specifies this schedule. The local proposition below instead refreshes the correction at every iteration.

He-Yang-Wang adaptation from either initialization reduces the median of the recorded scene-wise iteration ratios against the fixed structural setting. The opening column nevertheless requires more iterations with either initialization, and every adaptive method is capped on the dense random system although the fixed rule converges. Raw iteration medians order the initializations differently from paired ratios, and the normalized structural variant has the largest convergence count. These paired ratios support adaptation under the stated refresh schedule; the shift ablation below tests the dependence of that result on the schedule. No initialization is preferred independently of the evaluation metric. A run stopped at its cap remains a failure to attain the target. Its count is a lower bound on its own convergence work, but a ratio of two capped counts is not in general a lower bound on their convergence-time ratio.

4.2 Dependence on the de Saxcé shift

Refreshing the shift is necessary for coverage of this benchmark: removing it leaves only eight scenes reaching the corrected residual target. The benchmark contains twenty-one scenes. The grid-reference penalty reaches the target on every scene with the refreshed schedule. Removing the shift solves the associated convex subproblem, but the stopping test still evaluates the corrected Coulomb residual. Failure here therefore means failure to reach that target, not necessarily failure of the convex iteration to solve its own problem.

The following ablation holds the operator, initialization, linear solver, penalty rules, and stopping test fixed while comparing a refreshed correction with a removed or frozen correction. The frozen variant retains the first computed shift after the initial inner block; freezing the literal zero initialization instead is equivalent to removing the shift. Medians include capped counts, and penalty changes count sparse refactorizations across the benchmark.

Table 1, rows 1-8 of 12.

Penalty rule	Shift	Median recorded iterations [count]	Converged / tested [count]	Penalty changes [count]
Grid reference	Refreshed	278	21/21	0
Fixed structure	Refreshed	354	15/21	0
Smallest positive eigenvalue	Refreshed	1537	14/21	0
He, structure	Refreshed	376	16/21	368
Grid reference	Removed	20000	8/21	0
Fixed structure	Removed	20000	8/21	0
Smallest positive eigenvalue	Removed	20000	8/21	0
He, structure	Removed	20000	8/21	42795

Table 1, rows 9-12 of 12.

Penalty rule	Shift	Median recorded iterations [count]	Converged / tested [count]	Penalty changes [count]
Grid reference	Frozen after first block	20000	7/21	0
Fixed structure	Frozen after first block	30000	4/21	0
Smallest positive eigenvalue	Frozen after first block	30000	4/21	0
He, structure	Frozen after first block	30000	4/21	63045

Every rule reaches the target without the shift on eight scenes. The recorded ordering agrees with the refreshed schedule on seven of these scenes. The exception is the dense random contact system, whose outcomes are shown below. C denotes failure at the iteration cap. The refreshed comparison includes capped methods, so this is an ordering of recorded outcomes rather than of converged costs for every method.

Penalty rule	Refreshed [iterations]	Removed [iterations]	Frozen after first block [iterations]
Grid reference	1105	2529	20000 C
Fixed structure	3017	6864	20000 C
Smallest positive eigenvalue	20000 C	1377	20000 C
He, structure	20000 C	44	20000 C

On this scene, the refreshed and unshifted grid-reference runs both pass the corrected Coulomb residual test. Their stored reaction arrays differ at the byte level, but the records do not quantify the difference. They therefore do not establish distinct solution branches or nonuniqueness of the contact problem. The reversal is an outcome of the penalty rule together with the shift schedule and initialization. Residual balancing improves the paired aggregate for the refreshed implementation, but this ablation does not compare different refresh frequencies that all retain the full Coulomb correction. The ablation isolates this dependence but does not turn the dimensional inner-block threshold into a scale-invariant algorithm.

4.3 A local obstruction to a spectral rule

A sliding contact gives a direct explanation for the missing information. The correction depends on tangential velocity even if the Delassus matrix is isotropic. Consequently, the local iteration depends on friction and on the schedule used to refresh that correction.

Proposition 6 (local sliding rate).

Let $G=gI$ with positive g , use Euclidean cone projection without over-relaxation, and refresh $\Gamma(Gz+b)$ every iteration. Assume positive friction and penalty parameters. At a fixed point, write

$$\lambda=\ell(1,\mu e), u=(0,-ve), \|e\|=1.$$

The vector e is tangent to the contact plane. Here ℓ is the positive normal impulse and v is the positive slip speed. The linearized spectrum is

$$\{0,0,0, g/(g+\rho), (\rho+g\mu^2/(1+\mu^2))/(g+\rho), (\rho^2+gk)/((g+\rho)(\rho+k))\}, k=v/(\mu\ell). \\ \rho_{\text{opt}} = g/(1+\mu^2), q_{\text{opt}} = (1+\mu^2)/(2+\mu^2).$$

Proof. Write the scaled dual variable as d , and denote the de Saxcé correction by S . The differentiated map is

$$A=(G+\rho I)^{-1}, \\ x^+=A[\rho(z-d)-b-S(Gz+b)], \\ w=x^++d, z^+=\Pi_K\mu(w), d^+=w-z^+.$$

In the intermediate projection branch, define

$$w=(a, re), \ell=(a+\mu r)/(1+\mu^2), \\ q=(1,\mu e)/\sqrt{1+\mu^2}, c=(0, e_{\text{perp}}).$$

The transverse tangent is a unit vector orthogonal to e . Differentiation gives

$$P'=qq^T+\chi cc^T, \chi=\mu\ell/r=\rho/(\rho+k), k=v/(\mu\ell), \\ D=S'(u)G, B=A(\rho I-D), C=I-\rho A, \\ T\rho=[P'; I-P'][B, C], H=BP'+C(I-P').$$

The full derivative has the eigenvalues of H together with additional zeros. Choose the normal-tangent plane coordinates with tangent axis opposite to e , and define

$$b_1=(1,-\mu)/\sqrt{1+\mu^2}, b_2=(\mu,1)/\sqrt{1+\mu^2}, \\ a_\rho=\rho/(g+\rho), \beta=g/(g+\rho).$$

For the isotropic operator, substitution in this basis gives the triangular plane block

$$\begin{bmatrix} a_\rho+\beta\mu^2/(1+\mu^2) & 0 \\ \beta\mu^3/(1+\mu^2) & \beta \end{bmatrix}.$$

The transverse eigenvalue is $a_\rho\chi+\beta(1-\chi)$, a convex combination of a_ρ and β . The first plane eigenvalue dominates a_ρ , so the transverse branch cannot exceed the largest plane branch. The decreasing and increasing plane branches meet at the stated penalty, giving the unique spectral-radius minimum. With the correction frozen, the derivative term D vanishes and the plane eigenvalues are a_ρ and β . Their crossing is

$$\rho=g, q_{\text{frozen}}=1/2.$$

None of these derivatives applies across zero slip speed without separate directional analysis. ■

This proposition concerns the asymptotic rate near the specified sliding fixed point. It does not minimize cold-start iterations at a finite tolerance. Freezing the correction changes the differentiated map while leaving the Delassus operator unchanged. A bounded search did not locate the optimal constant in prior penalty-selection literature; this is a null result of that search,

not a priority claim. A spectral penalty rule that ignores the contact law and refresh schedule therefore cannot be deduced from the operator identity.

4.4 Stopping convention and refresh schedule

Two questions remain open from the local analysis: whether the stopping tolerance is tied to the chosen units, and whether the de Saxcé refresh frequency, with the contact law unchanged, dominates the iteration count. We measure both on the exported operators rather than through the local linearisation.

For the first, one physical problem per scene is re-expressed in five unit systems with length unit a metres and mass unit c kilograms (time unchanged), so $G \rightarrow cG$ and $b \rightarrow b/a$, and the impulse scales as $\lambda \rightarrow \lambda/(c a)$. Three stopping rules are compared. The delivered absolute rule $r < 1e-10$ is written in the current unit. The delivered normalised rule $r < 1e-10/(a c)$ applies the fixed physical tolerance $1e-10$ N s in every unit. The dimensionless rule $r/\lambda_{\text{ref}} < 1e-10$ uses the pure-number threshold with the impulse reference scale $\lambda_{\text{ref}} = \|b\|_{\infty} / \|G\|_{\infty}$. All three use the fixed structural penalty, the same zero initial state, and the full de Saxcé correction. The physical and dimensionless arms also share the dimensionless inner-block target, whereas the absolute arm uses an inner-block target written in the current unit; the absolute arm therefore changes both the outer stop and the inner target, and the comparison isolates the outer stopping convention only for the physical-versus-dimensionless contrast. The cap is 30000 iterations.

Table 3, panel 1/3, rows 1-8 of 9.

Scene	Stop	SI	Lp3	Lm3
Cube	Absolute $r < 1e-10$	75	51	97
Cube	Physical $1e-10$ N s	67	67	67
Cube	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	76	76	76
Eight-box tower	Absolute $r < 1e-10$	48	29	68
Eight-box tower	Physical $1e-10$ N s	48	48	48
Eight-box tower	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	70	70	70
Mass-ratio column, rest	Absolute $r < 1e-10$	42	32	30000 C
Mass-ratio column, rest	Physical $1e-10$ N s	42	42	42

Table 3, panel 1/3, rows 9-9 of 9.

Scene	Stop	SI	Lp3	Lm3
Mass-ratio column, rest	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	43	43	43

Table 3, panel 2/3, rows 1-8 of 9.

Scene	Stop	Mp3	Mm3	Distinct counts
Cube	Absolute $r < 1e-10$	40	294	5
Cube	Physical $1e-10$ N s	67	67	1
Cube	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	76	76	1
Eight-box tower	Absolute $r < 1e-10$	29	68	3
Eight-box tower	Physical $1e-10$ N s	48	48	1
Eight-box tower	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	70	70	1
Mass-ratio column, rest	Absolute $r < 1e-10$	32	51	4 (1 censored)
Mass-ratio column, rest	Physical $1e-10$ N s	42	42	1

Table 3, panel 2/3, rows 9-9 of 9.

Scene	Stop	Mp3	Mm3	Distinct counts
Mass-ratio column, rest	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	43	43	1

Table 3, panel 3/3, rows 1-8 of 9.

Scene	Stop	Max relative physical deviation
Cube	Absolute $r < 1e-10$	$5.61e-07$
Cube	Physical $1e-10$ N s	$3.90e-15$
Cube	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	$5.03e-15$
Eight-box tower	Absolute $r < 1e-10$	$3.70e-05$
Eight-box tower	Physical $1e-10$ N s	$1.45e-13$
Eight-box tower	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	$2.08e-13$
Mass-ratio column, rest	Absolute $r < 1e-10$	$2.81e-10$
Mass-ratio column, rest	Physical $1e-10$ N s	$4.76e-14$

Table 3, panel 3/3, rows 9-9 of 9.

Scene	Stop	Max relative physical deviation
Mass-ratio column, rest	Dimensionless $r/\lambda_{\text{ref}} < 1e-10$	$4.79e-14$

The physical and dimensionless stops both give one iteration count per scene across the five units, and their physical solutions agree to the stated relative deviation; the absolute rule gives three to five distinct counts per scene and its physical solution moves by up to $3.70e-05$. The mass-ratio column is censored in one unit under the absolute rule. The physical and dimensionless stops are therefore unit-invariant and the absolute stop is not. The physical and dimensionless stops differ from each other in value and in dimension; they are distinct algorithms, not two names for one rule. This is a bounded consistency check on three exported operator scenes, not a convergence theorem and not a speed claim, and it does not isolate adaptive-penalty behaviour, which the fixed penalty excludes.

The stored rows are reproduced from the packaged scene operator by the bundled solver. The dense-Cholesky path matches 44 of 45 outer iteration counts; the remaining cell, the mass-ratio column under the absolute rule at mass unit $1e-3$, is rounding-sensitive, dense Cholesky resting at its own residual floor while the sparse-LU path used in the measurement gives the stored count of 51 and 45 of 45 exact cells.

For the second question, the refresh schedule of the de Saxcé correction is varied while the contact law is held fixed. The arms are a per-iteration refresh, a between-inner-block refresh that

holds the correction fixed inside a block, and a between-block refresh with the dimensionless inner target. All three retain the full correction and the fixed structural penalty, so the comparison is same-law by construction; the third arm also changes the inner-block target, so it does not isolate refresh frequency alone. Cholesky factorisations number one for every arm.

Scene	Units	Per iteration	Between blocks	Between blocks, dimensionless target
Cube	5	57	75 (units 61 and 212)	67
Eight-box tower	5	48	48	48
Mass-ratio column, rest	5	42	42	42
Dense random contact system	3	3012	3017	3613
Packing, 50 grains	3	8000 C	8000 C	8000 C

Iteration counts change modestly on the well-conditioned operator and by about twenty percent on the dense random one, while Eight-box tower and the mass-ratio column are arm-insensitive. For this fixed structural penalty the refresh frequency is therefore not the dominant driver on these operators; the packing is censored in every arm, and the between-block arm is unit-dependent in two units even with the same law (61 and 212 against the SI/Lp3/Lm3 value 75). That unit effect traces to the dimensional inner-block refresh target used by this arm. All three refresh arms keep the physical outer termination test fixed. The third arm also changes the inner-block target, so comparisons involving it do not isolate refresh frequency alone. Because every arm uses a single factorisation, the cost separates through the measured factorisation-to-iteration ratio, which is scene-dependent; an adaptive rule that refactors cannot be ranked by iteration count alone. No universal arm ordering is claimed.

4.5 Relation to a splitting architecture for the same law

Song and colleagues [22] solve the same reduced Coulomb law $K_\mu \ni \lambda \perp u + \Gamma(u) \in K_\mu^*$ with the de Saxcé correction, citing the same cone-QP decomposition [7] that underlies the local Proposition 6. Their solver differs in three measurable ways. First, the outer map is a Tseng forward-backward-forward iteration, which evaluates the coupling, solves a proximal cone-constrained quadratic program, and applies an explicit projected correction; the measured solver here uses the Cadoux fixed-point outer loop with the correction refreshed between inner ADMM blocks, a Picard-type update. Second, they state no formal convergence rate for the non-monotone coupling and establish convergence empirically, which is the same regime our sweep reports as not uniformly predictable; they also report that globally coupled contact graphs lose contractivity. Third, they reject the natural-map residual as poorly scaled and replace it with a three-part dimensionless residual, whereas the measured tolerance here is a unit-consistent physical tolerance, itself not a dimensionless stop. Their outer parameter is an algorithmic step size derived from a Lipschitz bound on the coupling, not an ADMM penalty adapted by residual balance, so the two penalty-like quantities are not the same object. The block-sign and line-graph reduction of §2, the degree-three obstruction, and the patch-wrench quotient are not addressed there. That comparison is a positioning statement, not a novelty verdict; a bounded search is not a priority claim.

5. Load ordering from effective resistance

A scalar contact-graph surrogate can predict load ordering even when it does not reproduce the contact operator. Injecting load at the driven body and grounding the support gives a graph potential; the magnitude of its edge current is compared with contact-force magnitude. The conductance is inverse mass. This is a surrogate-flow experiment on eight scenes, not an assertion that each granular Delassus operator satisfies the block sign theorem.

Benchmark extent	Minimum [count]	Maximum [count]
Grains	50	63365
Contacts	176	124311

The global association is modest. Resistance flow improves on degree in the fine packings, and it recovers more of the largest loads in the gravity packing shown below. The rank statistic and top-set precision refer to different subsets of the benchmark; neither is an aggregate success rate over all scenes. These statistics are also solver- and tie-convention-sensitive, so they are reported as a heuristic ordering rather than a calibrated predictor. The delivered reference used an iterative solver at a loose tolerance; on the finest foot packing an independent direct solve moves the rank statistic by up to $5.2e-2$.

Predictor	Spearman ρ , fine packings [1]	Precision at top 10%, 50-grain packing [1]
Resistance flow	+0.390 ... +0.401	0.444
Contact degree	+0.322 ... +0.339	0.111
Distance to load	+0.110 ... +0.149	0.278

The advantage does not extend to local detection under the foot. Selecting the nearest contacts gives higher precision for the strongest-contact subset in the following cases.

Foot-loaded packing	Resistance-flow precision at top 5% [1]	Distance precision at top 5% [1]
990 grains	0.286	0.365
7921 grains	0.142	0.267

Changing conductance also changes the global ranking, so connectivity alone does not explain the result. On the finest packing the delivered iterative reference gives the measured Spearman values below; the corrected direct-solve recomputation moves the rank statistic by up to $5.2e-2$, as stated above.

Conductance	Spearman ρ , delivered iterative reference [1]
Inverse mass	+0.3898
Unweighted	+0.3651
Hertz stiffness	+0.3515

The rank convention contributes to this sensitivity. At a tie radius of $1e-9$ the leader-clustering rule reported here is compared with two alternatives for one foot packing.

Rank convention, corrected direct-solve reference, foot packing at tie radius $1e-9$	Spearman ρ [1]
Leader clustering (reported)	+0.3992709577045221
Transitive closure	+0.31899045477279486
Exact ties	+0.3996791630889775

The random-projection resistance sketch fails to preserve the directed-dipole ranking. Thus the evidence supports a load-ordering heuristic with modest global correlation, not a force reconstruction or a reliable local force-chain detector.

6. Contact-law checks and scope

Appendix G compares APGD and projected Gauss–Seidel for the convex relaxation with ADMM for the corrected Coulomb law. All recorded impulses pass the tangential inequality test, including the capped runs. That is compatible with projection onto the force cone and does not establish the original velocity complementarity condition. The cube shows a different normal-velocity diagnostic even when the relaxed solver reaches its own residual target. Conversely,

corrected Coulomb solutions can have positive normal velocities at separating contacts. The diagnostic is defined as

$$\text{gap} = \Delta t \max(0, \max_c u_{n,c}).$$

The step scale and tangential-inequality tolerance are in Appendix G. Because the statistic includes separating contacts, its maximum is not an estimate of artificial lift. The method-specific residual column and cap status must also be retained: a convex target and a corrected natural-map target are not interchangeable accuracy requirements. These measurements test the effect of the formulation and budget rather than a universal speed ordering.

Appendix H evaluates external friction-contact implementations on the same exported operators, biases, and friction coefficients. Siconos uses $u = Mr + q$, with $M = G_\eta$, $r = \lambda$, and $q = b$. The evaluated methods are nonsmooth Gauss–Seidel, the ACLM fixed point, proximal point, ADMM, de Saxcé fixed point, and nonsmooth Newton. The relative Siconos complementarity error is reported beside the absolute natural-map residual evaluated on the returned reaction. A run can pass the former test while exceeding the same numerical threshold in the latter. These are formulation checks, not matched-accuracy timing comparisons.

Different converged reactions can also arise from redundant force coordinates. The patch kernel identifies impulses invisible to body motion, but rank deficiency alone does not prove that any observed difference belongs to that kernel. Small residuals are necessary before interpreting a reaction as another admissible selection; capped, nonfinite, and missing results remain failures. The internal reference is itself a finite-tolerance solution. The stored cube and lattice references do not attain the absolute natural-map target used in the penalty-adaptation comparison. Relative reaction differences against them are therefore diagnostics, not independent error bounds on Siconos. A comparison of generalized body impulses at a common residual target is still needed to resolve force selection quantitatively.

The same distinction between operator and admissible reaction limits physical interpretation. At fixed geometry and inertia, changing load changes b without changing G . A reaction can reach the contact-cone boundary while the Delassus spectrum stays fixed. The spectrum can diagnose a difficult linear solve without locating a sliding or tipping threshold. In normalized patch coordinates and for positive supporting loads and friction, the tested cone margins take the form

$$\begin{aligned} m_1 &= \min_c (F_{n,c} - |M_{t1,c}|) / F_{n,c}, \\ m_2 &= \min_c (F_{n,c} - |M_{t2,c}|) / F_{n,c}, \\ m_f &= \min_c (\mu_c F_{n,c} - \|(F_{t1,c}, F_{t2,c}, M_{n,c})\|) / (\mu_c F_{n,c}), \\ m &= \min(m_1, m_2, m_f). \end{aligned}$$

A moment component alone is not this combined test, and the spin coordinate enters the friction norm. Normal-force positivity must be checked separately. These margins describe the chosen patch cone; they are not asserted to equal a ZMP margin or to predict a dynamic tipping event under arbitrary loading.

6.1 Quasistatic slip boundary

The static alternative is to retain all cone-feasible reactions and ask which applied loads they can support. For fixed contact geometry, let the nominal weight wrench be f_0 and let F map a horizontal acceleration into body force. For a unit load direction d , define

$$\begin{aligned} \mathcal{A} &= \{a: \exists \lambda \in K, J^T \lambda = f_0 + Fa\}, \\ r(d) &= \sup\{L \geq 0: Ld \in \mathcal{A}\}. \end{aligned}$$

This is the directional radius of a load section of the wrench cone. With circular Coulomb friction it need not be a polytope; polygonal friction gives a polyhedral representation, which is a polytope when the load section is bounded.

Proposition 7 (flat-support load radius). For the stated flat rectangular support model, equilibrium without sliding or loss of support requires both the friction bound and the centre-of-pressure bound. Let $a(d)$ be the distance from the nominal centre of pressure to the support boundary along the load ray, and let h be the centre-of-mass height. For a centred nominal centre of pressure, a rectangle with half-extents (e_1, e_2) and unit load direction $d=(dx, dy)$ has radial distance $\min(e_1/|dx|, e_2/|dy|)$ (omitting zero denominators). Its support function is $e_1|dx| + e_2|dy|$. These quantities generally differ; for example, they agree along the coordinate axes and toward a rectangle corner. For an off-centre nominal centre of pressure, $a(d)$ remains the distance to the boundary along the load ray. The directional limit is

$$r(d) = \min(\mu g, g a(d)/h).$$

Proof sketch. Horizontal force balance requires a tangential force of magnitude mL , while vertical balance gives normal force mg . Coulomb feasibility bounds their ratio. Moment balance displaces the centre of pressure by hL/g along the load direction, and the normal-cone image bounds this displacement by the ray distance from the nominal centre of pressure to the support boundary, i.e. the radial function $a(d)$, not the support function. For centred loading with no imposed yaw moment, sufficiency follows by choosing nonnegative normal loads with the required centre of pressure and distributing the tangential load in proportion to them. Each local Coulomb inequality then follows from the resultant friction bound. Their intersection gives the displayed radius. For stacked supports, the construction applies interface by interface using the supported substack and its centre of mass. ■

The ramp producer evaluates $a(d)$ in this ray form: for a rectangle it computes $g/h * \min((e_1 - \text{sign}(\cos)*u_0)/|\cos|, (e_2 - \text{sign}(\sin)*v_0)/|\sin|)$, which is the radial distance from the nominal centre of pressure, not the rectangle's support function. The ramp experiment detects displacement caused by sliding on the tested branch. Agreement with the static radius improves as the ramp slows, but the measured threshold includes the time needed to accumulate the prescribed displacement. The ramp applies a horizontal acceleration that increases linearly from zero to its trial peak. The simulation classifies a trial by relative or ground-referenced horizontal displacement, or by tilt; the reported boundary is reached by displacement first. The retained protocol is:

Protocol quantity	Value	Unit
Time step	0.005	s
PGS budget per step	60	sweeps
Contact residual target	$1e-8$	N s
Displacement threshold	5	mm
Tilt threshold	5	degrees
Initial peak acceleration	0.25	$m s^{-2}$
Bracketing growth factor	2	1
Bisections	8	count

The table reports mean relative error of the measured bracket midpoint against the static radius:

$$E = 100 |L_{\text{bracket}} - r|/r.$$

The saved wrench-radius table is indexed by geometry direction, with the source-row index preserved in a separate field. The longest ramps were tested only in the box's selected direction, so they do not validate all geometries or the tipping branch.

Scene / directions	Ramp time [s]	Relative threshold error [%]
Box, stack and pallet / 72 scene-direction pairs	0.05	442.16
Box, stack and pallet / 72 scene-direction pairs	0.2	94.23
Box, stack and pallet / 72 scene-direction pairs	1	24.03
Box, stack and pallet / 72 scene-direction pairs	5	7.49
Box, stack and pallet / 72 scene-direction pairs	20	2.73
Box / direction zero	50	1.458970
Box / direction zero	100	0.821865

The stack alone misses the combined experiment’s acceptance threshold at the slowest common ramp. Contact residuals also exceed the requested tolerance on some steps, so these are measured brackets rather than certified dynamic limits.

Ramp experiment limitation	Value	Unit
Stack error at slowest common ramp	3.051733	%
Acceptance threshold	3	%
Maximum final contact residual	1.5929817836033734e−04	N s
Requested residual tolerance	1e−08	N s

A force-closed grasp additionally needs normal-force limits for this radius to describe a finite clamp capacity. A frozen flat contact model also does not justify transfer to rolling or curved supports. These restrictions distinguish the measured quasistatic sliding result from a universal stability certificate.

6.2 Computational scope

All retained timings are CPU measurements on a shared host. The multigrid comparison includes setup; the preconditioner comparison measures the stated batch of solves after construction. No GPU speed claim is made. Repeatability checks apply only to runs for which paired outputs exist. They do not resolve differences in contact law or stopping residual.

6.3 Reproduction and aggregation

The operator problems for the penalty and contact-law comparisons are supplied as JSON and compact HDF5 and are reconstructed by the accompanying scene loader (`data/ncp/loader.py`). The patch and chain generator reproduces the delivered operators bit-identically. The load-ordering contact graphs and converged loads and the ramp bracket midpoints are exported outputs of the delivered runs, not regenerated grain-scale physics. The load-ordering rankings are recomputed on an independent direct reference with the fixed tie rule. The resistance-sketch metrics, the Appendix B and C iteration and multigrid counts, and the long-horizon ramp rows are aggregated historical data and are not recomputed. The nested stopping tolerance is a physical tolerance in the stated SI convention; it is not a fixed dimensionless stop, and §4.4 measures both it and a genuinely dimensionless stop on the same operators, three scenes across five unit systems. The dimensionless parts of the scheme are the cumulative-update condition and the residual-balancing decision, and the residual norms in that decision are two-norms. The delivered scene package, the patch-wrench, load-ordering and ramp supplement, and the dimensionless-stop comparison with its bundled solver are separate components of one bundle; each CSV in the tabulated studies is either regenerated from stored row-level measurements or delivered as a labelled aggregate, and the incidence-solver regeneration script writes all three of its CSVs.

7. Conclusion

Inverse spatial inertia gives an exact bodywise representation of contact coupling. Its reduction to a block Laplacian with a possibly zero Dirichlet term requires common coordinates, unique body pairs, and the incidence sign condition. Patch lumping removes internal point-force self-stress, while its admissible tangential wrench remains a constitutive choice. These properties justify chain elimination and explain why a scalar surrogate can fail on a packing. They do not select a universal nonlinear penalty: adaptation reduces the median paired ratio of recorded counts under the measured shift schedule but can fail where the fixed setting converges. The ablation reverses the ordering on the dense random system; the available reaction comparison does not resolve whether the runs select different solution branches. Graph structure is consequently a tool for specifying and solving the linear part of contact, whose usefulness must be assessed together with projection, compliance, and the stopping residual.

Appendix A. Assembly, rank, and spectral correspondence

The lattice retains the assembly and pencil identity while patch reduction leaves global rank deficiency.

Table 13, panel 1/2.

Scene	Patches [count]	Bodies of degree >2 [count]	Point rank/rows [count]
Three-box column	3	0	18/36
Eight-box tower	8	0	48/96
Mass-ratio column (1000:1)	3	0	18/36
Box lattice (2×2×2)	16	8	48/192

Table 13, panel 2/2.

Scene	Patch rank/rows [count]	Assembly relative ∞ -error [1]	Pencil relative spectral error [1]
Three-box column	18/18	2.368e−17	6.510e−16
Eight-box tower	48/48	9.586e−17	9.565e−16
Mass-ratio column (1000:1)	18/18	2.368e−17	4.836e−16
Box lattice (2×2×2)	48/96	2.368e−17	1.115e−15

Appendix B. Dense and block elimination

The final row compares unconverged iterates and does not measure solution error.

Table 14, panel 1/2.

Scene	Dense updates [count]	Block updates [count]	max abs impulse difference [N s]
Eight-box tower, gravity	140	140	1.67e−16
Eight-box tower, rolling	1848	1848	2.37e−15
Mass-ratio column (1000:1), gravity	296	296	4.39e−12
Mass-ratio column (1000:1), rolling	>30000	>30000	1.22e+01

Table 14, panel 2/2.

Scene	Status	Natural-map target [N s]
Eight-box tower, gravity	converged	1e-8
Eight-box tower, rolling	converged	1e-8
Mass-ratio column (1000:1), gravity	converged	1e-8
Mass-ratio column (1000:1), rolling	both capped; iterate difference	1e-8

Appendix C. Scalar incidence linear systems

Setup-inclusive CPU time and attained relative residual distinguish converged linear solves from stalled standalone cycles on $A = S \text{ diag}(1/m) S^T + \rho I$.

Ruge-Stüben + CG

Conjugate-gradient acceleration reaches the target on every listed incidence system.

Table 15, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	34
Mass-ratio patch incidence	3	0	2
Eight-body patch incidence	8	0	2
Asymmetric patch incidence	8	0	2
Mass-ratio point chain	3	0	2
Eight-body point chain	8	0	2
Granular step	441	0.358098622	93
Packing, 200 grains	710	2.864788976	108

Table 15, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	80
Packing, 800 grains	3022	2.864788976	130
Asymmetric tower incidence	32	0.01001	30

Table 15, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	0.003703660099	1.39391005e-11	pass
Mass-ratio patch incidence	0.0001848880202	1.697824553e-13	pass
Eight-body patch incidence	0.0001980180386	8.712091731e-15	pass
Asymmetric patch incidence	0.0002546079922	2.686039817e-12	pass
Mass-ratio point chain	0.000187835074	1.228302313e-13	pass
Eight-body point chain	0.0001932290616	8.712091731e-15	pass
Granular step	0.02097494493	8.859897596e-11	pass
Packing, 200 grains	0.03051988408	6.781539484e-11	pass

Table 15, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.01271233603	5.479209081e-11	pass
Packing, 800 grains	0.1069332911	5.3348256e-11	pass
Asymmetric tower incidence	0.003410444013	2.329227715e-11	pass

Ruge-Stüben V-cycle

Standalone cycles miss the target on the branching point-contact and packing cases.

Table 16, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	201
Mass-ratio patch incidence	3	0	2
Eight-body patch incidence	8	0	2
Asymmetric patch incidence	8	0	2
Mass-ratio point chain	3	0	2
Eight-body point chain	8	0	2
Granular step	441	0.358098622	201
Packing, 200 grains	710	2.864788976	201

Table 16, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	201
Packing, 800 grains	3022	2.864788976	201
Asymmetric tower incidence	32	0.01001	201

Table 16, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	0.009372626082	0.07636040734	target missed
Mass-ratio patch incidence	8.222402539e-05	2.619951042e-13	pass
Eight-body patch incidence	8.537597023e-05	1.082399872e-14	pass
Asymmetric patch incidence	8.883082774e-05	3.020610497e-12	pass
Mass-ratio point chain	8.282402996e-05	9.445502438e-14	pass
Eight-body point chain	8.821592201e-05	1.082399872e-14	pass
Granular step	0.03159748495	0.04891895978	target missed
Packing, 200 grains	0.04080992204	0.05939238848	target missed

Table 16, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.01927015395	0.03547710012	target missed
Packing, 800 grains	0.131928546	0.06840721127	target missed
Asymmetric tower incidence	0.009541906882	0.2012310665	target missed

Smoothed aggregation + CG

Conjugate-gradient acceleration reaches the target, with more iterations on packings than on chains.

Table 17, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	31
Mass-ratio patch incidence	3	0	2
Eight-body patch incidence	8	0	2
Asymmetric patch incidence	8	0	2
Mass-ratio point chain	3	0	2
Eight-body point chain	8	0	2
Granular step	441	0.358098622	128
Packing, 200 grains	710	2.864788976	141

Table 17, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	103
Packing, 800 grains	3022	2.864788976	160
Asymmetric tower incidence	32	0.01001	32

Table 17, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	0.004100079997	5.511167299e-12	pass
Mass-ratio patch incidence	0.0002522680443	1.697824553e-13	pass
Eight-body patch incidence	0.0002377199708	8.712091731e-15	pass
Asymmetric patch incidence	0.0002507490572	2.686039817e-12	pass
Mass-ratio point chain	0.0002478290116	1.228302313e-13	pass
Eight-body point chain	0.0002561140573	8.712091731e-15	pass
Granular step	0.02600767615	7.399822361e-11	pass
Packing, 200 grains	0.03725839104	7.544355978e-11	pass

Table 17, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.01779325691	3.833834424e-11	pass
Packing, 800 grains	0.1279786841	9.398701155e-11	pass
Asymmetric tower incidence	0.004393324023	4.394401455e-12	pass

Smoothed aggregation V-cycle

The standalone cycle reaches the target on chains and stalls on the other listed systems.

Table 18, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	201
Mass-ratio patch incidence	3	0	2
Eight-body patch incidence	8	0	2
Asymmetric patch incidence	8	0	2
Mass-ratio point chain	3	0	2
Eight-body point chain	8	0	2
Granular step	441	0.358098622	201
Packing, 200 grains	710	2.864788976	201

Table 18, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	201
Packing, 800 grains	3022	2.864788976	201
Asymmetric tower incidence	32	0.01001	201

Table 18, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	0.01025248098	0.1194322982	target missed
Mass-ratio patch incidence	0.0001476858743	2.619951042e-13	pass
Eight-body patch incidence	0.0001562670805	1.082399872e-14	pass
Asymmetric patch incidence	0.0001603679266	3.020610497e-12	pass
Mass-ratio point chain	0.0001517349156	9.445502438e-14	pass
Eight-body point chain	0.0001542770769	1.082399872e-14	pass
Granular step	0.02781967295	0.06168650859	target missed
Packing, 200 grains	0.03804021899	0.07377174546	target missed

Table 18, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.02202152892	0.04828668837	target missed
Packing, 800 grains	0.1277840671	0.0864185493	target missed
Asymmetric tower incidence	0.01050093805	0.07075948385	target missed

Direct

Direct solves reach the target on every listed incidence system.

Table 19, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	1
Mass-ratio patch incidence	3	0	1
Eight-body patch incidence	8	0	1
Asymmetric patch incidence	8	0	1
Mass-ratio point chain	3	0	1
Eight-body point chain	8	0	1
Granular step	441	0.358098622	1
Packing, 200 grains	710	2.864788976	1

Table 19, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	1
Packing, 800 grains	3022	2.864788976	1
Asymmetric tower incidence	32	0.01001	1

Table 19, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	7.706589531e-05	1.311443007e-14	pass
Mass-ratio patch incidence	1.878698822e-05	1.162189216e-13	pass
Eight-body patch incidence	1.735088881e-05	1.899354946e-15	pass
Asymmetric patch incidence	2.294604201e-05	8.703065953e-13	pass
Mass-ratio point chain	1.912808511e-05	1.133448696e-13	pass
Eight-body point chain	1.739000436e-05	1.899354946e-15	pass
Granular step	0.001556144911	5.148442586e-14	pass
Packing, 200 grains	0.003820893122	4.521111564e-14	pass

Table 19, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.0001962130191	2.397510841e-14	pass
Packing, 800 grains	0.03917622194	1.983819478e-14	pass
Asymmetric tower incidence	3.126298543e-05	2.731240243e-14	pass

Jacobi CG

Jacobi-preconditioned CG reaches the target without a multigrid hierarchy.

Table 20, panel 1/2, rows 1-8 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Eight-box tower incidence	32	0.02	10
Mass-ratio patch incidence	3	0	3
Eight-body patch incidence	8	0	8
Asymmetric patch incidence	8	0	8
Mass-ratio point chain	3	0	3
Eight-body point chain	8	0	8
Granular step	441	0.358098622	97
Packing, 200 grains	710	2.864788976	115

Table 20, panel 1/2, rows 9-11 of 11.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Packing, 50 grains	176	2.864788976	69
Packing, 800 grains	3022	2.864788976	152
Asymmetric tower incidence	32	0.01001	11

Table 20, panel 2/2, rows 1-8 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	0.0001135709463	7.474580249e-11	pass
Mass-ratio patch incidence	4.09099739e-05	6.465091218e-13	pass
Eight-body patch incidence	8.581799921e-05	2.415574774e-15	pass
Asymmetric patch incidence	8.863897528e-05	6.88798532e-13	pass
Mass-ratio point chain	4.1479012e-05	7.79024578e-13	pass
Eight-body point chain	8.650007658e-05	2.415574774e-15	pass
Granular step	0.001286617946	6.076249474e-11	pass
Packing, 200 grains	0.001801106031	7.394944173e-11	pass

Table 20, panel 2/2, rows 9-11 of 11.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 50 grains	0.0007753010141	3.160264002e-11	pass
Packing, 800 grains	0.007260465994	9.292384405e-11	pass
Asymmetric tower incidence	0.0001225449378	9.733895466e-14	pass

Tridiagonal Cholesky

The chain systems admit tridiagonal factorization and reach the target.

Table 21, panel 1/2.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]
Mass-ratio patch incidence	3	0	1
Eight-body patch incidence	8	0	1
Asymmetric patch incidence	8	0	1
Mass-ratio point chain	3	0	1
Eight-body point chain	8	0	1

Table 21, panel 2/2.

Scene	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Mass-ratio patch incidence	9.471084923e-06	1.162189216e-13	pass
Eight-body patch incidence	1.657998655e-05	1.899354946e-15	pass
Asymmetric patch incidence	1.723202877e-05	8.14012696e-13	pass
Mass-ratio point chain	9.119976312e-06	1.133448696e-13	pass
Eight-body point chain	1.666997559e-05	1.899354946e-15	pass

Appendix D. Preconditioned contact linear solves

Each scene uses identical right-hand sides across methods, with CPU times excluding preconditioner construction.

Graph

The graph preconditioner reduces iterations on the column but requires more iterations and solve time on the packing than the other methods.

Table 22, panel 1/2.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]
Mass-ratio column	0.002420391178	1	0.002859290107
Packing, 50 grains	469.2924618	62	2.71804603

Table 22, panel 2/2.

Scene	Max relative residual [1], target 1e-10	max abs(L-G _{nn}) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	9.25577049e-14	0	100
Packing, 50 grains	9.807603499e-11	286.4727554	100

Block Jacobi

On the packing, block Jacobi costs less solve time than the graph preconditioner.

Table 23, panel 1/2.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]
Mass-ratio column	0.002420391178	3	0.003835485084
Packing, 50 grains	469.2924618	35	0.20877871

Table 23, panel 2/2.

Scene	Max relative residual [1], target $1e-10$	max abs(L-G _{nn}) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	1.002617841e-13	0	100
Packing, 50 grains	9.253055396e-11	286.4727554	100

None

The unpreconditioned packing solve costs less time than either tested preconditioner.

Table 24, panel 1/2.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]
Mass-ratio column	0.002420391178	3	0.003077984089
Packing, 50 grains	469.2924618	33	0.1690009242

Table 24, panel 2/2.

Scene	Max relative residual [1], target $1e-10$	max abs(L-G _{nn}) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	6.764498941e-11	0	100
Packing, 50 grains	7.108833378e-11	286.4727554	100

Appendix E. Spectral penalties and grid minima

The parameter distance is $\max(\rho/\rho_{struct}, \rho_{struct}/\rho)$ and capped or nonfinite candidates remain in the accounting.

Table 25, panel 1/4, rows 1-8 of 21.

Scene	Contacts [count]	$\lambda_{\min}^+$ [kg ⁻¹]	$\lambda_{\max}$ [kg ⁻¹]
Cube	4	0.111696312	0.9549703547
Eight-box tower	32	0.00233314724	46.37933351
Mass-ratio column, rest	3	0.000292768252	2.00100025
Mass-ratio column, sliding	3	0.000292768252	2.00100025
Mass-ratio column, opening	3	0.000292768252	2.00100025
Asymmetric tower	32	3.77871314e-06	24.02119446
Box lattice	64	0.4021119685	65.15994714
Random contact network	256	0.2163405649	25.76148555

Table 25, panel 1/4, rows 9-16 of 21.

Scene	Contacts [count]	$\lambda_{\min}^+$ [kg ⁻¹]	$\lambda_{\max}$ [kg ⁻¹]
Granular step	441	0.6172496689	526.1260007
Packing, 50 grains	176	4.935228954	4306.254112
Packing, 200 grains	710	4.870694009	4386.170694
Packing, 800 grains	3022	4.773906827	4759.045125
Dense random contact system	256	126.7338501	2889.305391
Column, heavy top	3	0.000292768252	2.00100025
Eight-body constrained column	8	7.602904289e-05	2.001707419
Quadruped, sticking	4	0.01683484815	5.586013882

Table 25, panel 1/4, rows 17-21 of 21.

Scene	Contacts [count]	$\lambda_{\min}^+$ [kg ⁻¹]	$\lambda_{\max}$ [kg ⁻¹]
Quadruped, sliding	4	0.01683484815	5.586013882
Humanoid	2	0.05780543824	0.5933695941
Quadruped, unloading probe	4	0.2721286926	7.610089619
Quadruped, batch sticking state	4	0.01898863344	5.586100305
Quadruped, batch sliding state	4	0.01756848743	5.586491561

Table 25, panel 2/4, rows 1-8 of 21.

Scene	ρ^* [kg ⁻¹]	Iterations [count]	Rule
Cube	0.1986272518	58	B
Eight-box tower	2.33314724e-06	2	B
Mass-ratio column, rest	2.92768252e-07	4	A
Mass-ratio column, sliding	0.0009258145028	162	A
Mass-ratio column, opening	0.001646356868	27	A
Asymmetric tower	3.77871314e-09	2	B
Box lattice	0.0004021119685	2	B
Random contact network	3.847139722	202	B

Table 25, panel 2/4, rows 9-16 of 21.

Scene	ρ^* [kg ⁻¹]	Iterations [count]	Rule
Granular step	61.72496689	1104	B
Packing, 50 grains	493.5228954	577	B
Packing, 200 grains	487.0694009	833	B
Packing, 800 grains	477.3906827	926	B
Dense random contact system	225.3681962	800	B
Column, heavy top	0.01646356868	2681	A
Eight-body constrained column	0.01352008815	3027	A
Quadruped, sticking	1.683484815e-05	2	B

Table 25, panel 2/4, rows 17-21 of 21.

Scene	ρ^* [kg ⁻¹]	Iterations [count]	Rule
Quadruped, sliding	0.946693082	627	B
Humanoid	5.780543824e-05	3	B
Quadruped, unloading probe	2.721286926	56	B
Quadruped, batch sticking state	0.1898863344	200	B
Quadruped, batch sliding state	0.5555643533	248	B

Table 25, panel 3/4, rows 1-8 of 21.

Scene	Parameter distance [1]	Boundary	Capped [count]
Cube	1.644279068	false	6
Eight-box tower	140990.931	true	0
Mass-ratio column, rest	1000	true	3
Mass-ratio column, sliding	3.16227766	false	11
Mass-ratio column, opening	5.623413252	false	5
Asymmetric tower	2521304.725	true	1
Box lattice	12729.66169	true	0
Random contact network	1.629609426	false	12

Table 25, panel 3/4, rows 9-16 of 21.

Scene	Parameter distance [1]	Boundary	Capped [count]
Granular step	3.425196964	false	22
Packing, 50 grains	3.385351949	false	21
Packing, 200 grains	3.332365726	false	16
Packing, 800 grains	3.167211443	false	16
Dense random contact system	2.685037834	false	15
Column, heavy top	56.23413252	false	22
Eight-body constrained column	177.827941	false	23
Quadruped, sticking	18215.72258	true	0

Table 25, panel 3/4, rows 17-21 of 21.

Scene	Parameter distance [1]	Boundary	Capped [count]
Quadruped, sliding	3.087120605	false	15
Humanoid	3203.89515	true	3
Quadruped, unloading probe	1.891003279	false	8
Quadruped, batch sticking state	1.715171333	false	10
Quadruped, batch sliding state	1.773362876	false	14

Table 25, panel 4/4, rows 1-8 of 21.

Scene	Nonfinite [count]	$\eta_c/\ G\mathbf{0}_{cc}\ ^2$ [1]
Cube	0	0
Eight-box tower	0	0
Mass-ratio column, rest	0	0
Mass-ratio column, sliding	0	0
Mass-ratio column, opening	0	0
Asymmetric tower	0	0
Box lattice	0	0
Random contact network	0	0

Table 25, panel 4/4, rows 9-16 of 21.

Scene	Nonfinite [count]	$\eta_c/\ G\mathbf{0}_{cc}\ ^2$ [1]
Granular step	0	0.01
Packing, 50 grains	0	0.01
Packing, 200 grains	5	0.01
Packing, 800 grains	6	0.01
Dense random contact system	6	0
Column, heavy top	0	0
Eight-body constrained column	0	0
Quadruped, sticking	0	0

Table 25, panel 4/4, rows 17-21 of 21.

Scene	Nonfinite [count]	$\eta_c/\ G\mathbf{0}_{cc}\ ^2$ [1]
Quadruped, sliding	0	0
Humanoid	0	0
Quadruped, unloading probe	0	0
Quadruped, batch sticking state	0	0
Quadruped, batch sliding state	0	0

$\lambda_{\min}^+ > 10^{-11} \lambda_{\max}$, $(a_{\min}, a_{\max}, N_{\text{grid}}) = (-3, 3, 25)$, $R_\eta < 10^{-8} \text{ N s}$, $I_{\max} = 5000$.

Appendix F. Penalty adaptation with a common residual

C marks a capped run; every method uses the same operator, cold start, and natural-map target, including the rerun grid-reference penalty.

Table 26, panel 1/3, rows 1-8 of 21.

Scene	Cap [iterations]	Grid reference [iterations]	Fixed structure [iterations]
Cube	30000	70	75
Eight-box tower	30000	2	48
Mass-ratio column, rest	30000	5	42
Mass-ratio column, sliding	30000	198	260
Mass-ratio column, opening	30000	35	36
Asymmetric tower	30000	2	25811
Box lattice	30000	3	68
Random contact network	20000	281	354

Table 26, panel 1/3, rows 9-16 of 21.

Scene	Cap [iterations]	Grid reference [iterations]	Fixed structure [iterations]
Granular step	20000	1672	20000 C
Packing, 50 grains	30000	1180	30000 C
Packing, 200 grains	20000	1398	20000 C
Packing, 800 grains	2000	1499	2000 C
Dense random contact system	20000	1105	3017
Column, heavy top	30000	3282	30000 C
Eight-body constrained column	30000	4813	30000 C
Quadruped, sticking	30000	2	135

Table 26, panel 1/3, rows 17-21 of 21.

Scene	Cap [iterations]	Grid reference [iterations]	Fixed structure [iterations]
Quadruped, sliding	30000	807	1020
Humanoid	30000	3	80
Quadruped, unloading probe	30000	92	156
Quadruped, batch sticking state	30000	278	293
Quadruped, batch sliding state	30000	311	501

Table 26, panel 2/3, rows 1-8 of 21.

Scene	He, mean [iterations]	He, structure [iterations]	Normalized, mean [iterations]
Cube	69	75	143
Eight-box tower	13	9	13
Mass-ratio column, rest	21	10	21
Mass-ratio column, sliding	908	818	238
Mass-ratio column, opening	1878	376	47
Asymmetric tower	206	103	41
Box lattice	17	9	30000 C
Random contact network	284	354	284

Table 26, panel 2/3, rows 9-16 of 21.

Scene	He, mean [iterations]	He, structure [iterations]	Normalized, mean [iterations]
Granular step	20000 C	20000 C	20000 C
Packing, 50 grains	30000 C	30000 C	30000 C
Packing, 200 grains	20000 C	20000 C	20000 C
Packing, 800 grains	2000 C	2000 C	2000 C
Dense random contact system	20000 C	20000 C	20000 C
Column, heavy top	690	1858	955
Eight-body constrained column	5103	5762	30000 C
Quadruped, sticking	13	9	13

Table 26, panel 2/3, rows 17-21 of 21.

Scene	He, mean [iterations]	He, structure [iterations]	Normalized, mean [iterations]
Quadruped, sliding	683	695	1102
Humanoid	12	11	12
Quadruped, unloading probe	330	302	195
Quadruped, batch sticking state	195	249	1254
Quadruped, batch sliding state	464	443	1382

Table 26, panel 3/3, rows 1-8 of 21.

Scene	Normalized, structure [iterations]
Cube	118
Eight-box tower	9
Mass-ratio column, rest	10
Mass-ratio column, sliding	513
Mass-ratio column, opening	36
Asymmetric tower	18
Box lattice	9
Random contact network	354

Table 26, panel 3/3, rows 9-16 of 21.

Scene	Normalized, structure [iterations]
Granular step	20000 C
Packing, 50 grains	1378
Packing, 200 grains	20000 C
Packing, 800 grains	2000 C
Dense random contact system	20000 C
Column, heavy top	1373
Eight-body constrained column	4494
Quadruped, sticking	9

Table 26, panel 3/3, rows 17-21 of 21.

Scene	Normalized, structure [iterations]
Quadruped, sliding	1386
Humanoid	11
Quadruped, unloading probe	253
Quadruped, batch sticking state	1146
Quadruped, batch sliding state	773

$$R_{\eta} < 10^{-10} \text{ N s}, \quad \tau = 2, \quad \kappa = 10, \\ \|r_p\|_2 > \kappa \|r_d\|_2 = \rho \leftarrow \tau \rho, \quad \|r_d\|_2 > \kappa \|r_p\|_2 = \rho \leftarrow \rho / \tau.$$

The inner block holds the current tangential-speed vector fixed and refreshes it after the primal residual falls below the following numerical threshold:

$$\begin{aligned}
s^{(0)} &= 0, \quad s_c^{(j+1)} = \|(G_\eta z + b)_{t,c}\|_2, \\
\delta_0 &= \max(\epsilon, 0.1 \|b\|_\infty), \\
\delta_j &= \max(\epsilon, 0.1 \max(\|s^{(j)} - s^{(j-1)}\|_\infty, \epsilon)), \quad j > 0, \\
\|x - z\|_2 &< \delta_j, \quad \text{total inner updates} \geq 5, \quad \epsilon = 10^{-10}.
\end{aligned}$$

This inner refresh criterion compares numerical values of an impulse residual ($\|x - z\|_2$) with velocity quantities, including the tangential-speed change ($\|s^{(j)} - s^{(j-1)}\|_\infty$). The implementation uses the numerical coefficient 0.1 without a declared physical conversion factor. We report this as the raw SI criterion; it is not a dimensionally consistent, unit-invariant tolerance. Section 4.4 separately evaluates a dimensionless inner target and distinguishes the physical outer termination test from a normalized outer test. The total-update condition is cumulative across blocks. The corrected natural-map residual is checked at every inner update, before adaptation or shift refresh, and remains the termination criterion. The unscaled dual variable is preserved across penalty changes. These details distinguish this measured nested iteration from the per-iteration refresh used in the local proposition.

Appendix G. Convex and corrected contact laws

Residuals belong to each method’s own law; the normal-gap diagnostic includes separating contacts.

APGD

APGD can converge to its relaxed-law target while retaining a positive normal-gap diagnostic.

Table 27, panel 1/2, rows 1-8 of 21.

Scene	Iterations [count]	Cap [count]	Status
Cube	101	5000	converged
Eight-box tower	2562	5000	converged
Mass-ratio column, rest	5000	5000	capped
Mass-ratio column, sliding	5000	5000	capped
Mass-ratio column, opening	5000	5000	capped
Asymmetric tower	5000	5000	capped
Box lattice	576	5000	converged
Random contact network	3079	5000	converged

Table 27, panel 1/2, rows 9-16 of 21.

Scene	Iterations [count]	Cap [count]	Status
Granular step	5000	5000	capped
Packing, 50 grains	3895	5000	converged
Packing, 200 grains	5000	5000	capped
Packing, 800 grains	5000	5000	capped
Dense random contact system	1221	5000	converged
Column, heavy top	5000	5000	capped
Eight-body constrained column	5000	5000	capped
Quadruped, sticking	468	5000	converged

Table 27, panel 1/2, rows 17-21 of 21.

Scene	Iterations [count]	Cap [count]	Status
Quadruped, sliding	817	5000	converged
Humanoid	229	5000	converged
Quadruped, unloading probe	402	5000	converged
Quadruped, batch sticking state	842	5000	converged
Quadruped, batch sliding state	818	5000	converged

Table 27, panel 2/2, rows 1-8 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	6.43026743e-11	3.270000014e-05	0
Eight-box tower	9.777877119e-11	2.240763651e-12	0
Mass-ratio column, rest	0.06362969869	2.651237446e-07	0
Mass-ratio column, sliding	0.05286465984	0.0004161165714	0
Mass-ratio column, opening	0.0001746373258	0.0004169794296	0
Asymmetric tower	2.627224926e-05	3.281364609e-08	0
Box lattice	6.206488448e-11	5.4977753e-20	0
Random contact network	8.262893147e-11	0.001516901924	0

Table 27, panel 2/2, rows 9-16 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Granular step	9.343661489e-09	0.01291150628	0
Packing, 50 grains	9.950401516e-11	0.005844353264	0
Packing, 200 grains	3.011538213e-10	0.005998695167	0
Packing, 800 grains	9.46212492e-10	0.005989463739	0
Dense random contact system	9.775151522e-11	1.35097248e-10	0
Column, heavy top	0.07635563843	0.0008333580174	0
Eight-body constrained column	0.4616320989	0.0008334576305	0
Quadruped, sticking	9.738644613e-11	2.266645089e-12	0

Table 27, panel 2/2, rows 17-21 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Quadruped, sliding	2.782746256e-11	5.286511295e-06	0
Humanoid	6.667943975e-11	0	0
Quadruped, unloading probe	6.5659575e-11	8.001231629e-05	0
Quadruped, batch sticking state	7.506481547e-11	3.733374691e-06	0
Quadruped, batch sliding state	3.688355177e-12	1.223284581e-05	0

Convex PGS

The scene-dependent budgets leave several projected Gauss-Seidel runs above their relaxed-law target.

Table 28, panel 1/2, rows 1-8 of 21.

Scene	Iterations [count]	Cap [count]	Status
Cube	53	5000	converged
Eight-box tower	4500	4500	capped
Mass-ratio column, rest	5000	5000	capped
Mass-ratio column, sliding	5000	5000	capped
Mass-ratio column, opening	5000	5000	capped
Asymmetric tower	4500	4500	capped
Box lattice	261	5000	converged
Random contact network	644	1000	converged

Table 28, panel 1/2, rows 9-16 of 21.

Scene	Iterations [count]	Cap [count]	Status
Granular step	400	400	capped
Packing, 50 grains	913	1000	converged
Packing, 200 grains	250	250	capped
Packing, 800 grains	60	60	capped
Dense random contact system	550	550	capped
Column, heavy top	5000	5000	capped
Eight-body constrained column	5000	5000	capped
Quadruped, sticking	2490	5000	converged

Table 28, panel 1/2, rows 17-21 of 21.

Scene	Iterations [count]	Cap [count]	Status
Quadruped, sliding	1341	5000	converged
Humanoid	198	5000	converged
Quadruped, unloading probe	418	5000	converged
Quadruped, batch sticking state	1310	5000	converged
Quadruped, batch sliding state	1146	5000	converged

Table 28, panel 2/2, rows 1-8 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	8.949960858e-11	3.270000003e-05	0
Eight-box tower	2.519047193e-06	5.772816483e-08	0
Mass-ratio column, rest	0.006788612785	0	0
Mass-ratio column, sliding	0.00678513913	0.000416032746	0
Mass-ratio column, opening	4.562734586e-05	0.0004178030123	0
Asymmetric tower	0.002165968144	0	0
Box lattice	9.881346782e-11	5.314010724e-12	0
Random contact network	9.923387112e-11	0.001516901924	0

Table 28, panel 2/2, rows 9-16 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Granular step	1.47431094e-05	0.01291050442	0
Packing, 50 grains	9.886788437e-11	0.005844353266	0
Packing, 200 grains	1.347454571e-06	0.005998770029	0
Packing, 800 grains	3.315655711e-05	0.00598916608	0
Dense random contact system	6.715880001e-06	4.058504171e-05	0
Column, heavy top	0.008146335342	0.0008333333333	0
Eight-body constrained column	0.04037576716	0.0008314102311	0
Quadruped, sticking	9.983275318e-11	2.281182332e-12	0

Table 28, panel 2/2, rows 17-21 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Quadruped, sliding	9.984302274e-11	5.286507372e-06	0
Humanoid	9.80975301e-11	0	0
Quadruped, unloading probe	9.713191362e-11	8.001230962e-05	0
Quadruped, batch sticking state	9.851050531e-11	3.733367731e-06	0
Quadruped, batch sliding state	9.879524976e-11	1.22328407e-05	0

Coulomb ADMM

The corrected solver reaches its target in every listed scene; positive normal gaps remain possible at separating contacts.

Table 29, panel 1/2, rows 1-8 of 21.

Scene	Iterations [count]	Cap [count]	Status
Cube	70	5000	converged
Eight-box tower	2	5000	converged
Mass-ratio column, rest	5	5000	converged
Mass-ratio column, sliding	198	5000	converged
Mass-ratio column, opening	35	5000	converged
Asymmetric tower	2	5000	converged
Box lattice	3	5000	converged
Random contact network	281	5000	converged

Table 29, panel 1/2, rows 9-16 of 21.

Scene	Iterations [count]	Cap [count]	Status
Granular step	1672	5000	converged
Packing, 50 grains	1180	5000	converged
Packing, 200 grains	1398	5000	converged
Packing, 800 grains	1499	5000	converged
Dense random contact system	1105	5000	converged
Column, heavy top	3282	5000	converged
Eight-body constrained column	4813	5000	converged
Quadruped, sticking	2	5000	converged

Table 29, panel 1/2, rows 17-21 of 21.

Scene	Iterations [count]	Cap [count]	Status
Quadruped, sliding	807	5000	converged
Humanoid	3	5000	converged
Quadruped, unloading probe	92	5000	converged
Quadruped, batch sticking state	278	5000	converged
Quadruped, batch sliding state	311	5000	converged

Table 29, panel 2/2, rows 1-8 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	7.450456918e-11	1.422830892e-13	0
Eight-box tower	4.951733468e-13	0	0
Mass-ratio column, rest	2.273736754e-13	4.736951572e-16	0
Mass-ratio column, sliding	8.446932043e-11	4.736951572e-16	0
Mass-ratio column, opening	7.668177204e-11	0.0004177078125	0
Asymmetric tower	4.170885859e-12	8.21037104e-14	0
Box lattice	1.085208312e-13	1.275469654e-15	0
Random contact network	9.019551425e-11	0.001688735757	0

Table 29, panel 2/2, rows 9-16 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Granular step	9.982629307e-11	0.01320401497	0
Packing, 50 grains	9.947796102e-11	0.005844240581	0
Packing, 200 grains	9.938339279e-11	0.006096702174	0
Packing, 800 grains	9.985446237e-11	0.005989526298	0
Dense random contact system	9.991044971e-11	0.02370284443	0
Column, heavy top	9.958966984e-11	0	0
Eight-body constrained column	9.965783754e-11	0	0
Quadruped, sticking	2.059552529e-11	0	0

Table 29, panel 2/2, rows 17-21 of 21.

Scene	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Quadruped, sliding	8.900271709e-11	2.582387891e-12	0
Humanoid	6.562261845e-11	0	0
Quadruped, unloading probe	2.403774488e-11	0	0
Quadruped, batch sticking state	9.285461549e-11	2.285389181e-13	0
Quadruped, batch sliding state	6.128820368e-11	0	0

$\Delta t = 1/240$ s, tangential test fails if $\|\lambda_t\| > \mu\lambda_n + 10^{-12}$ N s.

Appendix H. External friction-contact solutions

Siconos relative stopping errors and absolute natural-map residuals are shown separately, with no matched-accuracy speed inference.

NSGS

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Table 30, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]
Cube	converged	28	3.119242636e-11	7.080810072e-12
Eight-box tower	converged	36138	9.994128621e-11	5.98493477e-13
Mass-ratio column, rest	converged	23061	9.981710826e-11	9.993073036e-11
Mass-ratio column, sliding	converged	22526	8.510786222e-11	2.252136255e-10
Mass-ratio column, opening	converged	11443	9.987405774e-11	1.163584784e-10
Asymmetric tower	iteration cap	200000	8.063739006e-05	1.667788212e-05
Box lattice	converged	201	9.7574752e-11	3.969533036e-13
Random contact network	converged	122	2.862340751e-11	2.352779083e-11

Table 30, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]
Granular step	converged	482	9.909073602e-11	1.632661194e-11
Packing, 50 grains	converged	557	2.747599854e-11	6.188151472e-14
Packing, 200 grains	converged	696	4.811111969e-11	4.975932318e-13
Packing, 800 grains	converged	724	9.043927436e-11	5.928003314e-13
Dense random contact system	converged	576	6.625714818e-11	8.813059657e-12
Column, heavy top	converged	22602	9.994367934e-11	1.265334504e-10
Eight-body constrained column	converged	73028	5.56297353e-11	1.045660225e-10
Quadruped, sticking	converged	132	4.870892392e-11	4.181724411e-13

Table 30, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	converged	92	1.267880645e-11	4.226619055e-13
Humanoid	converged	6	9.815478326e-14	8.205242041e-15
Quadruped, unloading probe	converged	6	2.518981392e-15	3.122502257e-17
Quadruped, batch sticking state	converged	43	3.093408589e-11	3.750055821e-13
Quadruped, batch sliding state	converged	71	8.273748574e-12	2.959195389e-13

Table 30, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	0.05590169703
Eight-box tower	converged	0.2220458214
Mass-ratio column, rest	converged	9.917666169e-11
Mass-ratio column, sliding	converged	2.050074307e-10
Mass-ratio column, opening	converged	5.353683565e-09
Asymmetric tower	iteration cap	0.5048144963
Box lattice	converged	0.4111027742
Random contact network	converged	4.249420871e-07

Table 30, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	converged	4.699232181e-06
Packing, 50 grains	converged	0.0008072604959
Packing, 200 grains	converged	0.0001995416797
Packing, 800 grains	converged	7.287681841e-05
Dense random contact system	converged	4.147401393e-06
Column, heavy top	converged	1.241471842e-10
Eight-body constrained column	converged	4.429070464e-07
Quadruped, sticking	converged	2.906827478e-09

Table 30, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	converged	8.52151708e-06
Humanoid	converged	9.958433509e-10
Quadruped, unloading probe	converged	1.872049997e-06
Quadruped, batch sticking state	converged	6.215054932e-07
Quadruped, batch sliding state	converged	2.735310156e-06

ACLMFP

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Table 31, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Cube	converged	66	8.782108299e-11	1.132571814e-11
Eight-box tower	converged	47408	9.998550962e-11	5.463268726e-13
Mass-ratio column, rest	converged	29471	9.998225769e-11	6.069490155e-08
Mass-ratio column, sliding	converged	17497	9.99801098e-11	9.989616956e-08
Mass-ratio column, opening	converged	10846	9.998990048e-11	1.081730261e-10
Asymmetric tower	iteration cap	100000	6.61512121e-05	1.15881481e-05
Box lattice	converged	257	9.873123625e-11	3.368407983e-13
Random contact network	converged	158	9.331661175e-11	3.662745801e-11

Table 31, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Granular step	converged	676	9.967782436e-11	2.162775167e-11
Packing, 50 grains	iteration cap	100000	2.434586967e-08	5.779904724e-11
Packing, 200 grains	wall limit or error	—	—	—
Packing, 800 grains	wall limit or error	—	—	—
Dense random contact system	wall limit or error	—	—	—
Column, heavy top	converged	24805	9.996339867e-11	7.550249848e-08
Eight-body constrained column	converged	85977	9.999657395e-11	1.073743594e-07
Quadruped, sticking	converged	1068	9.958796911e-11	7.861489237e-13

Table 31, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	converged	687	9.815923189e-11	2.751368577e-12
Humanoid	converged	141	9.949239809e-11	7.692957382e-12
Quadruped, unloading probe	converged	31	7.646929485e-11	5.260653024e-13
Quadruped, batch sticking state	converged	233	9.556573987e-11	1.290204055e-12
Quadruped, batch sliding state	converged	605	9.532804496e-11	3.17216392e-12

Table 31, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	0.1218080142
Eight-box tower	converged	0.2214364408
Mass-ratio column, rest	converged	1.856154512e-10
Mass-ratio column, sliding	converged	4.195114329e-10
Mass-ratio column, opening	converged	5.388817994e-09
Asymmetric tower	iteration cap	0.4946032172
Box lattice	converged	0.4071982673
Random contact network	converged	4.266287422e-07

Table 31, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	converged	4.690862821e-06
Packing, 50 grains	iteration cap	0.0008068265272
Packing, 200 grains	wall limit or error	—
Packing, 800 grains	wall limit or error	—
Dense random contact system	wall limit or error	—
Column, heavy top	converged	2.464333506e-10
Eight-body constrained column	converged	1.563755988e-06
Quadruped, sticking	converged	3.155308899e-09

Table 31, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	converged	8.519220228e-06
Humanoid	converged	8.919651816e-10
Quadruped, unloading probe	converged	1.872087784e-06
Quadruped, batch sticking state	converged	6.212439137e-07
Quadruped, batch sliding state	converged	2.732522416e-06

PROX

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Table 32, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Cube	converged	21	1.195048333e-15	2.498001805e-16
Eight-box tower	converged	26	1.187564286e-11	4.485301019e-14
Mass-ratio column, rest	converged	8910	1.72414626e-12	1.317857823e-09
Mass-ratio column, sliding	converged	4567	6.669023245e-12	7.414200809e-09
Mass-ratio column, opening	converged	3044	5.680060299e-11	4.234834705e-11
Asymmetric tower	converged	443	8.064423322e-13	1.207922651e-13
Box lattice	converged	17	7.483188261e-15	3.304411454e-16
Random contact network	converged	7	3.536199358e-16	2.220444801e-16

Table 32, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Granular step	converged	4	9.179025474e-14	6.537218683e-14
Packing, 50 grains	converged	59	3.816544388e-12	2.902707998e-13
Packing, 200 grains	wall limit or error	—	—	—
Packing, 800 grains	wall limit or error	—	—	—
Dense random contact system	converged	5	3.817844118e-12	3.80141179e-11
Column, heavy top	converged	5640	1.271057486e-13	2.155502443e-10
Eight-body constrained column	converged	22	1.710280014e-11	2.192547377e-08
Quadruped, sticking	converged	22	1.1288416e-12	8.54871729e-15

Table 32, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	converged	14	3.736126891e-15	2.892651396e-16
Humanoid	converged	92	6.001882487e-17	8.881784197e-16
Quadruped, unloading probe	converged	15	8.235519967e-16	1.387778781e-17
Quadruped, batch sticking state	converged	25	4.052399307e-11	5.354050536e-13
Quadruped, batch sliding state	converged	11	3.391772795e-11	2.072163188e-12

Table 32, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	5.052887144e-08
Eight-box tower	converged	1.000224453e-06
Mass-ratio column, rest	converged	2.941058437e-11
Mass-ratio column, sliding	converged	6.309629157e-11
Mass-ratio column, opening	converged	5.322042327e-09
Asymmetric tower	converged	0.011978239
Box lattice	converged	2.943114736e-07
Random contact network	converged	4.251110231e-07

Table 32, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	converged	4.696746953e-06
Packing, 50 grains	converged	0.0008072591948
Packing, 200 grains	wall limit or error	—
Packing, 800 grains	wall limit or error	—
Dense random contact system	converged	4.145632619e-06
Column, heavy top	converged	3.220425029e-11
Eight-body constrained column	converged	5.364301665e-07
Quadruped, sticking	converged	2.90556666e-09

Table 32, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	converged	8.521804576e-06
Humanoid	converged	9.959136974e-10
Quadruped, unloading probe	converged	1.872049998e-06
Quadruped, batch sticking state	converged	6.214240505e-07
Quadruped, batch sliding state	converged	2.734630354e-06

ADMM

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Table 33, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Cube	converged	100	1.99962858e-12	3.853861674e-13
Eight-box tower	converged	43	8.838426188e-11	3.071015664e-13
Mass-ratio column, rest	converged	15213	3.351371511e-11	2.376748398e-08
Mass-ratio column, sliding	converged	145713	5.492617119e-11	6.31102921e-08
Mass-ratio column, opening	converged	465	9.554086874e-11	7.878497854e-11
Asymmetric tower	iteration cap	200000	2.233042998e-05	2.479208142e-06
Box lattice	converged	26	5.146269863e-12	3.123890036e-14
Random contact network	converged	231	2.025274095e-11	3.128269573e-11

Table 33, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Granular step	converged	1339	1.328375051e-11	2.005329609e-11
Packing, 50 grains	converged	3478	2.810092043e-11	9.556434207e-12
Packing, 200 grains	converged	4785	1.603306024e-12	8.371752456e-13
Packing, 800 grains	converged	4585	1.282413998e-11	1.122244946e-11
Dense random contact system	wall limit or error	—	—	—
Column, heavy top	converged	101083	9.996463504e-11	8.752579106e-08
Eight-body constrained column	iteration cap	200000	2.486150043e-08	3.351768052e-05
Quadruped, sticking	converged	125	3.584456863e-12	2.57294186e-14

Table 33, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	converged	1055	5.972642016e-11	2.404035304e-12
Humanoid	converged	113	5.799374575e-12	4.151123889e-13
Quadruped, unloading probe	converged	172	2.008777701e-12	5.851569229e-14
Quadruped, batch sticking state	converged	224	3.886356208e-12	1.293635338e-13
Quadruped, batch sliding state	converged	1152	1.944582277e-15	3.365363543e-16

Table 33, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	5.052298146e-08
Eight-box tower	converged	3.212830341e-10
Mass-ratio column, rest	converged	6.695662359e-11
Mass-ratio column, sliding	converged	2.54125747e-10
Mass-ratio column, opening	converged	5.39477999e-09
Asymmetric tower	iteration cap	0.07748569217
Box lattice	converged	6.895668638e-08
Random contact network	converged	4.250891628e-07

Table 33, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	converged	4.696756437e-06
Packing, 50 grains	converged	0.0008072571709
Packing, 200 grains	converged	0.000199541461
Packing, 800 grains	converged	7.287686058e-05
Dense random contact system	wall limit or error	—
Column, heavy top	converged	2.693286516e-10
Eight-body constrained column	iteration cap	0.0003651861153
Quadruped, sticking	converged	2.900985165e-09

Table 33, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	converged	8.520896583e-06
Humanoid	converged	9.898142908e-10
Quadruped, unloading probe	converged	1.872055084e-06
Quadruped, batch sticking state	converged	6.215954419e-07
Quadruped, batch sliding state	converged	2.735523256e-06

DSFP

The de Saxcé fixed-point method includes nonfinite outputs and capped runs; these are failures, irrespective of the reported iteration count.

Table 34, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Cube	converged	97	8.885339433e-11	1.667976868e-11
Eight-box tower	iteration cap	200000	2.82236756e+87	1.118521162e+85
Mass-ratio column, rest	iteration cap	200000	1.013473563e+21	7.163376343e+20
Mass-ratio column, sliding	iteration cap	200000	71.29839568	90.83869789
Mass-ratio column, opening	iteration cap	200000	4.73494208e+20	3.902918378e+20
Asymmetric tower	iteration cap	200000	4.320894993e+153	3.323632051e+153
Box lattice	iteration cap	200000	580.2490844	7.783789091
Random contact network	iteration cap	200000	nonfinite	—

Table 34, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Granular step	iteration cap	200000	nonfinite	—
Packing, 50 grains	iteration cap	200000	2.02350991e+105	1.251296802e+103
Packing, 200 grains	iteration cap	200000	5.245513264e+38	4.477906324e+36
Packing, 800 grains	wall limit or error	—	—	—
Dense random contact system	wall limit or error	—	—	—
Column, heavy top	converged	93380	9.998800196e-11	8.950667052e-08
Eight-body constrained column	iteration cap	200000	2.527415886e-05	5.135840246e-05
Quadruped, sticking	iteration cap	200000	1.158875492	0.02391664236

Table 34, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	iteration cap	200000	0.33817247	0.02534091624
Humanoid	converged	405	9.971855857e-11	8.047562616e-12
Quadruped, unloading probe	iteration cap	200000	1	0.005372673136
Quadruped, batch sticking state	iteration cap	200000	0.9465417767	0.03907821554
Quadruped, batch sliding state	iteration cap	200000	0.2633533011	0.02668787057

Table 34, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	5.065120362e-08
Eight-box tower	iteration cap	7.852524632e+85
Mass-ratio column, rest	iteration cap	4.477921096e+17
Mass-ratio column, sliding	iteration cap	0.919838375
Mass-ratio column, opening	iteration cap	5.9761805e+17
Asymmetric tower	iteration cap	5.589988934e+151
Box lattice	iteration cap	251.1975491
Random contact network	iteration cap	nonfinite

Table 34, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	iteration cap	nonfinite
Packing, 50 grains	iteration cap	8.202543881e+105
Packing, 200 grains	iteration cap	1.078227256e+39
Packing, 800 grains	wall limit or error	—
Dense random contact system	wall limit or error	—
Column, heavy top	converged	2.773120989e-10
Eight-body constrained column	iteration cap	0.0006414031528
Quadruped, sticking	iteration cap	0.6543405324

Table 34, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	iteration cap	0.7600255177
Humanoid	converged	8.92613923e-10
Quadruped, unloading probe	iteration cap	1
Quadruped, batch sticking state	iteration cap	0.7399587151
Quadruped, batch sliding state	iteration cap	0.7235440118

NSN AC

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Table 35, panel 1/2, rows 1-8 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Cube	converged	1	2.502018505e-16	1.592316813e-17
Eight-box tower	converged	730	1.957576129e-12	2.011585343e-14
Mass-ratio column, rest	converged	3	1.607774678e-13	2.273736754e-13
Mass-ratio column, sliding	converged	4	1.200666826e-13	1.705302566e-13
Mass-ratio column, opening	converged	2	9.748565909e-14	1.136868377e-13
Asymmetric tower	iteration cap	10001	0.01973507073	0.006133655368
Box lattice	converged	2814	1.266132896e-11	8.619702174e-13
Random contact network	converged	7	4.354035828e-11	1.031931103e-10

Table 35, panel 1/2, rows 9-16 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Granular step	converged	13	7.220457066e-15	3.229447959e-14
Packing, 50 grains	converged	14	1.984925267e-11	5.188776756e-12
Packing, 200 grains	wall limit or error	—	—	—
Packing, 800 grains	wall limit or error	—	—	—
Dense random contact system	converged	14	2.299109643e-11	1.399806469e-10
Column, heavy top	converged	3	1.44922851e-13	2.273736754e-13
Eight-body constrained column	converged	26	1.812986607e-16	2.442490654e-15
Quadruped, sticking	converged	1	6.656433011e-16	6.938893904e-18

Table 35, panel 1/2, rows 17-21 of 21.

Scene	Status	Iterations [count]	Siconos residual [1], target 1e-10	Natural-map residual [N s]
Quadruped, sliding	converged	4	1.230955805e-11	1.894063899e-12
Humanoid	converged	1	2.400752995e-16	2.775557562e-17
Quadruped, unloading probe	converged	4	2.541362653e-13	2.102484853e-15
Quadruped, batch sticking state	converged	6	3.709804386e-16	1.734723476e-18
Quadruped, batch sliding state	converged	5	4.35430705e-12	5.480802444e-13

Table 35, panel 2/2, rows 1-8 of 21.

Scene	Status	Relative impulse difference [1]
Cube	converged	0.4472135764
Eight-box tower	converged	1.131419213
Mass-ratio column, rest	converged	1.128825108e-12
Mass-ratio column, sliding	converged	2.565752091e-11
Mass-ratio column, opening	converged	5.237091862e-09
Asymmetric tower	iteration cap	0.8306568771
Box lattice	converged	1.190006091
Random contact network	converged	4.25157156e-07

Table 35, panel 2/2, rows 9-16 of 21.

Scene	Status	Relative impulse difference [1]
Granular step	converged	4.696747562e-06
Packing, 50 grains	converged	0.000807261616
Packing, 200 grains	wall limit or error	—
Packing, 800 grains	wall limit or error	—
Dense random contact system	converged	4.147557684e-06
Column, heavy top	converged	3.477711282e-11
Eight-body constrained column	converged	7.221080274e-07
Quadruped, sticking	converged	2.907839753e-09

Table 35, panel 2/2, rows 17-21 of 21.

Scene	Status	Relative impulse difference [1]
Quadruped, sliding	converged	8.521901905e-06
Humanoid	converged	9.959044269e-10
Quadruped, unloading probe	converged	1.872049774e-06
Quadruped, batch sticking state	converged	6.215954638e-07
Quadruped, batch sliding state	converged	2.735553634e-06

Appendix I. Scene definitions

Each scene used in the penalty and contact-law comparisons is supplied as JSON and compact HDF5 in the accompanying supplement’s `data/ncp/scenes/` directory, with reconstruction and verification in `data/ncp/loader.py`. The file stem in the table identifies the matching JSON and HDF5 files. The scene manifest lists the files and their checksums. For the cube, the pair is

```
data/ncp/scenes/00.json
data/ncp/scenes/00.hdf5
```

JSON stores the bias, friction, time step, compliance, and either body/contact geometry or the complete generalized operator. HDF5 stores the resulting contact problem. This scene dataset is distinct from the contact-graph reproduction supplement, which covers the patch-wrench, load-ordering and ramp experiments.

The eigenvalues refer to the operator used in the penalty experiment, including its declared compliance. The condition number is the ratio of the largest eigenvalue to the smallest retained positive eigenvalue, rather than a claim of nonsingularity. The cutoff is the one in Appendix E.

Table 36, panel 1/3, rows 1-8 of 21.

File stem	Scene	n_b [count]	n_c [count]	Maximum degree [count]
00	Cube	1	4	4
01	Eight-box tower	8	32	8
02	Mass-ratio column, rest	3	3	2
03	Mass-ratio column, sliding	3	3	2
04	Mass-ratio column, opening	3	3	2
05	Asymmetric tower	8	32	8
06	Box lattice	8	64	16
07	Random contact network	64	256	13

Table 36, panel 1/3, rows 9-16 of 21.

File stem	Scene	n_b [count]	n_c [count]	Maximum degree [count]
08	Granular step	125	441	14
09	Packing, 50 grains	50	176	10
10	Packing, 200 grains	200	710	10
11	Packing, 800 grains	800	3022	11
12	Dense random contact system	32	256	256
13	Column, heavy top	3	3	2
14	Eight-body constrained column	8	8	2
15	Quadruped, sticking	1	4	4

Table 36, panel 1/3, rows 17-21 of 21.

File stem	Scene	n_b [count]	n_c [count]	Maximum degree [count]
16	Quadruped, sliding	1	4	4
17	Humanoid	1	2	2
18	Quadruped, unloading probe	1	4	4
19	Quadruped, batch sticking state	1	4	4
20	Quadruped, batch sliding state	1	4	4

Table 36, panel 2/3, rows 1-8 of 21.

File stem	Scene	Mass ratio / operator ratio [1]	μ interval [1]	$\lambda_{\min}^+$ [kg ⁻¹]
00	Cube	1	[0.5, 0.5]	0.111696
01	Eight-box tower	1	[0.5, 0.5]	0.00233315
02	Mass-ratio column, rest	1000	[0.5, 0.5]	0.000292768
03	Mass-ratio column, sliding	1000	[0.5, 0.5]	0.000292768
04	Mass-ratio column, opening	1000	[0.5, 0.5]	0.000292768
05	Asymmetric tower	1000	[0.5, 0.5]	3.77871e-06
06	Box lattice	1	[0.5, 0.5]	0.402112
07	Random contact network	1	[0.5, 0.5]	0.216341

Table 36, panel 2/3, rows 9-16 of 21.

File stem	Scene	Mass ratio / operator ratio [1]	μ interval [1]	$\lambda_{\min}^+$ [kg ⁻¹]
08	Granular step	1	[0.4, 0.5]	0.61725
09	Packing, 50 grains	1	[0.5, 0.5]	4.93523
10	Packing, 200 grains	1	[0.5, 0.5]	4.87069
11	Packing, 800 grains	1	[0.5, 0.5]	4.77391
12	Dense random contact system	7.23226	[0.35, 0.35]	126.734
13	Column, heavy top	1000	[0.5, 0.5]	0.000292768
14	Eight-body constrained column	1000	[0.5, 0.5]	7.6029e-05
15	Quadruped, sticking	5458.96	[0.5, 0.5]	0.0168348

Table 36, panel 2/3, rows 17-21 of 21.

File stem	Scene	Mass ratio / operator ratio [1]	μ interval [1]	$\lambda_{\min}^+$ [kg ⁻¹]
16	Quadruped, sliding	5458.96	[0.5, 0.5]	0.0168348
17	Humanoid	9711.64	[0.5, 0.5]	0.0578054
18	Quadruped, unloading probe	3153.45	[0.5, 0.5]	0.272129
19	Quadruped, batch sticking state	5472.16	[0.5, 0.5]	0.0189886
20	Quadruped, batch sliding state	5465.08	[0.5, 0.5]	0.0175685

Table 36, panel 3/3, rows 1-8 of 21.

File stem	Scene	$\lambda_{\max}$ [kg ⁻¹]	κ^+ [1]
00	Cube	0.95497	8.5497
01	Eight-box tower	46.3793	19878.4
02	Mass-ratio column, rest	2.001	6834.76
03	Mass-ratio column, sliding	2.001	6834.76
04	Mass-ratio column, opening	2.001	6834.76
05	Asymmetric tower	24.0212	6.35698e+06
06	Box lattice	65.1599	162.044
07	Random contact network	25.7615	119.078

Table 36, panel 3/3, rows 9-16 of 21.

File stem	Scene	$\lambda_{\max}$ [kg ⁻¹]	κ^+ [1]
08	Granular step	526.126	852.371
09	Packing, 50 grains	4306.25	872.554
10	Packing, 200 grains	4386.17	900.523
11	Packing, 800 grains	4759.05	996.887
12	Dense random contact system	2889.31	22.7982
13	Column, heavy top	2.001	6834.76
14	Eight-body constrained column	2.00171	26328.2
15	Quadruped, sticking	5.58601	331.813

Table 36, panel 3/3, rows 17-21 of 21.

File stem	Scene	$\lambda_{\max}$ [kg ⁻¹]	κ^+ [1]
16	Quadruped, sliding	5.58601	331.813
17	Humanoid	0.59337	10.2649
18	Quadruped, unloading probe	7.61009	27.965
19	Quadruped, batch sticking state	5.5861	294.181
20	Quadruped, batch sliding state	5.58649	317.984

For geometric scenes, n_b counts dynamic bodies, degree counts incident point contacts, and mass ratio is the largest body mass divided by the smallest. Contact rows are ordered normal then tangents; body columns are translation then rotation when rotation is present. The JSON supplies contact directions, body pairs, and moment arms in the coordinates used to assemble each body block. The loader places the contact directions in the translational columns and their cross products with the moment arms in the rotational columns, with opposite signs for the incident bodies. Ground has no dynamic mass block.

For operator scenes, the exported body count and degree describe the generator’s bookkeeping, not an articulated robot’s physical link graph. The reported mass ratio is the ratio of largest to smallest positive reciprocal diagonal entries of the generalized inverse mass; it mixes generalized coordinates and is not a physical body-mass ratio. The dense random system has no underlying contact geometry. Robot scenes supply the full Jacobian and inverse mass in generalized velocity coordinates, so reconstruction does not require inferring moment arms from contact rows.

The compact HDF5 matrix stores the lower triangle in compressed sparse column form with the `symmetric_packed` attribute. The supplied reader restores its symmetric counterpart; an unmodified standard FCLIB reader must not treat the stored triangle as the full operator. Running `python data/ncp/loader.py` rebuilds the contact operator from each JSON and compares it with its HDF5 reference. This dataset defines the penalty and contact-law experiments. It does not contain the patch-wrench systems, the load-ordering packings, or the ramp trajectories used in the other experiments; those experiments cannot be reconstructed from this dataset alone.

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